

# Reciprocal-Tail Rigidity: Complete Reasoning Record

Erdős Problem #243

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## ABSTRACT

If strictly increasing positive integers satisfy  $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$ , then  $\sum_n 1/a_n$  is irrational (Theorem 2.1), and Section 2 proves it in full. Rationality clears denominators into an exact positive integer orbit  $C_{n+1} = a_n C_n - D_n$ ,  $D_{n+1} = a_n D_n$ ; the prescribed rate extracts the exact profile  $C_n = A n(n+1)(n+2) + B$ ; integrality forces the primitive shape  $\frac{m}{6}n(n+1)(n+2) + c$  with  $c = \pm 1$ ; the denominator transport forces  $\alpha^2 - 1$  to be a square in the cubic field generated by a root  $\alpha$  of  $T^3 - T + 6c/m$ ; a trace calculation leaves the single scale  $m = 12$ ; and the two survivors  $2n(n+1)(n+2) \pm 1$  carry four-step words modulo 7 that the exact recursion cannot realise. The rate class lies inside the growth hypothesis of Erdős Problem #243 and outside the two sufficient conditions recorded in the literature, and Problem #243 remains open outside that class. The rest of the note works under the hypotheses of the problem itself: a target-equivalent weighted record-excess criterion, further criteria in the record coordinates, exclusions of constant, periodic, bounded and finite-normalised-mass negative errors, two conditional endpoints on the integer state system, and the exact open boundary. The lead theorem is an ordinary proof with an exact finite replay of every finite claim, and no Lean declaration covers it; the statements marked Lean-checked are the state-system results, linked to the pinned source revision 92b88dc1bbe0.

## 1 Introduction

Sylvester's sequence 2, 3, 7, 43, 1807, ... is generated by  $a_n = a_{n-1}^2 - a_{n-1} + 1$  and has reciprocal sum 1; the recurrence is exactly the statement that the tail beyond  $a_{n-1}$  equals  $1/(a_{n-1} - 1)$ . Erdős Problem #243 asserts that this is the only way a sequence of that growth can have a rational reciprocal sum:

**Problem 1.1** (Erdős #243). Let  $1 \leq a_1 < a_2 < \dots$  be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then  $a_n = a_{n-1}^2 - a_{n-1} + 1$  for all sufficiently large  $n$ .

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\**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

See [4, p. 64] and [5, p. 105]. Bloom's catalogue records the problem as open [8]. Below we index the same recurrence as  $a_{n+1} = a_n^2 - a_n + 1$ , which is the shift the formal sources use; the two forms are the same statement. Write  $\text{syl}(a) = a^2 - a + 1$ , the [Sylvester successor](#).

The hypothesis is asymptotic and the conclusion is exact, so the argument must convert an analytic rate into an integer obstruction. The conversion used here is the classical one: clear denominators along the sequence, so that rationality makes a tail integral, and then centre that integer at the value it would take on Sylvester's sequence. What remains is a single integer error  $E$ , and the whole question is whether  $E$  can avoid vanishing.

Section 2 carries out that conversion at one prescribed rate and settles the assertion of Problem 1.1 on the resulting class of sequences. Section 3 places the theorem against the published sufficient conditions, states the evidence class of every result here, and guides the rest of the note.

**Keywords.** irrationality; Ahmes series; Sylvester's sequence; unit fractions; Lean 4. **MSC 2020.** 11J72 (primary); 11B37, 11D68, 68V20 (secondary).

## 2 An unconditional irrationality theorem

This section proves the note's lead theorem and the arithmetic exclusion it rests on. It is self-contained: the integer state it uses is constructed here in six lines, and Section 4 returns to that construction at leisure.

**Theorem 2.1** (cubic-rate irrationality). *If strictly increasing positive integers satisfy  $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$ , then  $\sum_n 1/a_n$  is irrational.*

**Theorem 2.2** (rising-factorial cubic exclusion). *Let  $a, C, D : \mathbb{N} \rightarrow \mathbb{Z}_{>0}$  satisfy*

$$C_{n+1} = a_n C_n - D_n, \quad D_{n+1} = a_n D_n. \quad (1)$$

*Then for every  $A \in \mathbb{Q}_{>0}$  and every  $B \in \mathbb{Q}$ ,*

$$\liminf_{X \rightarrow \infty} \frac{\#\{n \leq X : C_n \neq A n(n+1)(n+2) + B\}}{X} > 0.$$

*In particular no such orbit satisfies  $C_n = A n(n+1)(n+2) + B$  for all large  $n$ .*

The quantifier in Theorem 2.2 is the one that the irrationality theorem needs. It excludes agreement along prefixes whose proportion of exceptions tends to zero, along every subsequence of prefix lengths, which is more than the exclusion of eventual equality. The two restrictions that meet on the same cubic are integrality, which supplies the rising-factorial shape and the scale, and the denominator transport, which imposes an arithmetic square condition; the scale calculation and the two residue words that close the argument can be checked on the page.

**The canonical integer tail.** Suppose  $\sum_{n \geq 1} 1/a_n = p/q$  with  $p, q \geq 1$ , and write  $x_n = \sum_{k \geq n} 1/a_k$  for the tail. Put

$$D_n = q \prod_{k < n} a_k, \quad C_n = D_n x_n.$$

Then  $D_n$  is a positive integer, and

$$C_n = D_n \left( \frac{p}{q} - \sum_{k < n} \frac{1}{a_k} \right) = p \prod_{k < n} a_k - q \sum_{k < n} \prod_{\substack{j < n \\ j \neq k}} a_j$$

is a positive integer as well. From  $x_n = 1/a_n + x_{n+1}$ ,

$$C_{n+1} = D_{n+1} x_{n+1} = a_n D_n \left( x_n - \frac{1}{a_n} \right) = a_n C_n - D_n, \quad D_{n+1} = a_n D_n,$$

which is (1). Section 4 constructs the same triple with its centred error  $E_n = D_n - (a_n - 1)C_n$ , which the present section does not need. Rationality is used here and nowhere else below.

## 2.1 Sparse agreement forces a primitive integral cubic

For  $S \subseteq \mathbb{N}$  write  $\underline{d}(S) = \liminf_X \#(S \cap [1, X]) / X$ .

**Lemma 2.3** (periodic obstruction principle). *Let  $h \geq 1$ , let  $j_1, \dots, j_L$  be fixed offsets, and suppose that for every sufficiently large  $n$  in one residue class modulo  $h$  at least one of  $n + j_1, \dots, n + j_L$  lies in  $S$ . Then  $\underline{d}(S) \geq 1/(Lh)$ .*

*Proof.* There are  $X/h + O(1)$  relevant starting indices below  $X$ , and each element of  $S$  serves as a witness for at most  $L$  of them.  $\square$

Fix  $A \in \mathbb{Q}_{>0}$ ,  $B \in \mathbb{Q}$ , and suppose for contradiction that

$$P(n) = A n(n+1)(n+2) + B, \quad S = \{n : C_n \neq P(n)\}, \quad \underline{d}(S) = 0.$$

Everything through the end of Section 2.4 runs under that assumption.

**Lemma 2.4** (gcd stabilisation and the primitive shape). *Write  $G_n = \gcd(C_n, D_n)$ . Then  $6A$  is a positive integer,  $G_n$  divides  $6A$  for every  $n$ , and  $G_n$  is eventually equal to a positive integer  $g$ . On the tail where  $G_n = g$ , put  $u_n = C_n/g$ ,  $v_n = D_n/g$  and  $Q(n) = P(n)/g$ . Then*

$$u_{n+1} = a_n u_n - v_n, \quad v_{n+1} = a_n v_n, \quad \gcd(u_n, v_n) = 1, \quad \gcd(u_n, u_{n+1}) = 1, \quad (2)$$

and there are  $m \in \mathbb{Z}_{>0}$  and  $c \in \{-1, 1\}$  with

$$Q(n) = \frac{m}{6} n(n+1)(n+2) + c. \quad (3)$$

*Proof.* Equation (1) gives  $G_n \mid G_{n+1}$  and  $G_n \mid C_t$  for every  $t \geq n$ . Since  $\underline{d}(S) = 0$ , beyond every threshold there is a block of four consecutive indices disjoint from  $S$ : otherwise every block of four would meet  $S$  and  $\underline{d}(S) \geq 1/4$ . On such a block  $\Delta^3 C_t = \Delta^3 P(t) = 6A$ , and  $G_n$  divides each of the four values, hence divides  $6A$ . So  $6A$  is a positive integer and the divisibility chain  $(G_n)$  is bounded and stabilises at some  $g$ .

On the primitive tail,  $\gcd(u_n, v_n) = 1$  by construction, and a prime dividing  $u_n$  and  $u_{n+1}$  would divide  $v_n = a_n u_n - u_{n+1}$ , which gives  $\gcd(u_n, u_{n+1}) = 1$ .

The polynomial  $Q$  takes an integer value at every integer argument. Indeed, let  $N$  clear the denominators of its coefficients, so  $NQ \in \mathbb{Z}[X]$  and  $Q(n+N) - Q(n) \in \mathbb{Z}$  for every  $n$ . A non-integral value at one argument would therefore force an exception at every late argument in one residue class modulo  $N$ , and Lemma 2.3 with  $L = 1$  would contradict  $\underline{d}(S) = 0$ . Now  $Q(-1) = Q(0) = B/g$ , so  $c := B/g \in \mathbb{Z}$ , and  $m := Q(1) - Q(0) = 6A/g$  is a positive integer, which is (3) with  $c \in \mathbb{Z}$ .

It remains to exclude  $c = 0$  and every  $c$  with a prime factor. Let  $\ell$  be a prime dividing  $c$ , or any prime if  $c = 0$ . For  $n \equiv -1 \pmod{6\ell}$ , write  $n+1 = 6\ell k$ ; then

$$\frac{n(n+1)(n+2)}{6} = \ell k(6\ell k - 1)(6\ell k + 1), \quad \frac{(n+1)(n+2)(n+3)}{6} = \ell k(n+2)(n+3),$$

so  $\ell$  divides both  $Q(n)$  and  $Q(n+1)$ . Agreement at both indices would give  $\ell \mid \gcd(u_n, u_{n+1}) = 1$ . Hence one of  $n, n+1$  lies in  $S$  for every late  $n \equiv -1 \pmod{6\ell}$ , and Lemma 2.3 with  $L = 2, h = 6\ell$  gives  $\underline{d}(S) \geq 1/(12\ell)$ . This contradiction leaves  $c = \pm 1$ .  $\square$

**Lemma 2.5** (the multiplier supply, and the reducible case). *On the primitive tail,  $\gcd(a_n, v_n) = 1$ , the multipliers at distinct indices are pairwise coprime, infinitely many of them exceed 1, and the cubic  $Q_{m,c}$  of (3) is irreducible over  $\mathbb{Q}$ .*

*Proof.* A prime dividing  $a_n$  and  $v_n$  would divide  $u_{n+1}$  and  $v_{n+1}$ , against  $\gcd(u_{n+1}, v_{n+1}) = 1$ ; and every earlier multiplier divides every later  $v_n$ , so distinct multipliers are coprime. If  $a_n = 1$  for all large  $n$  then  $v_n$  is eventually a positive constant and  $u_{n+1} = u_n - v_n$  decreases without bound, against positivity. So infinitely many distinct primes divide late multipliers.

Suppose  $Q_{m,c}$  had a rational root. Choose a prime  $\ell \nmid 6m$  dividing a late multiplier  $a_N$ , large enough that the root reduces modulo  $\ell$ . Then  $Q_{m,c}$  vanishes modulo  $\ell$  on a full residue class, while  $\ell \mid v_n$  and therefore  $\ell \nmid u_n$  for every  $n > N$ . Agreement at any late index of that class is impossible, and Lemma 2.3 with  $L = 1, h = \ell$  gives  $\underline{d}(S) \geq 1/\ell$ .  $\square$

## 2.2 The square forced by the transport

**Lemma 2.6** (square specialisation). *Let  $f \in \mathbb{Q}[T]$  be irreducible with root  $\alpha$ , and let  $H \in \mathbb{Q}[T]$  satisfy  $H(\alpha) \neq 0$ . If for all but finitely many primes  $\ell$  every root  $r \in \mathbb{F}_\ell$  of the reduction of  $f$  has  $H(r)$  a square in  $\mathbb{F}_\ell^\times$ , then  $H(\alpha)$  is a square in  $\mathbb{Q}(\alpha)^\times$ .*

*Proof.* Suppose not. Put  $K = \mathbb{Q}(\alpha)$  and choose  $\beta$  with  $\beta^2 = H(\alpha)$ , so  $K(\beta)/K$  is quadratic. Take a finite Galois extension  $L/\mathbb{Q}$  containing  $K(\beta)$  and  $\sigma \in \text{Gal}(L/K)$  restricting non-trivially to  $K(\beta)$ , so that  $\sigma\alpha = \alpha$  and  $\sigma\beta = -\beta$ . By the Chebotarev density theorem there are infinitely many unramified rational primes whose Frobenius class in  $\text{Gal}(L/\mathbb{Q})$  is that of  $\sigma$ ; only the infinitude is used. Discard the finitely many primes dividing denominators or the discriminant, the prime 2, and those at which  $\beta$  reduces to zero. At a prime of  $L$  above such an  $\ell$  with Frobenius  $\sigma$ , the residue field satisfies  $\bar{\alpha}^\ell = \bar{\alpha}$  and  $\bar{\beta}^\ell = -\bar{\beta}$ , so  $\bar{\alpha} \in \mathbb{F}_\ell$  while  $\bar{\beta} \notin \mathbb{F}_\ell$ . Then  $H(\bar{\alpha}) = \bar{\beta}^2$  is a nonsquare in  $\mathbb{F}_\ell$  at a root of the reduction of  $f$ , against the hypothesis.  $\square$

The hypothesis is needed at every root and at all but finitely many primes. A favourable finite set of tested primes would not suffice.

**Proposition 2.7** (the transport square). *Write  $\kappa = m/6$  and  $\eta = 6c/m$ , so that  $Q_{m,c}(n) = \kappa f(n+1)$  for  $f(T) = T^3 - T + \eta$ . Then  $\alpha^2 - 1$  is a square in  $\mathbb{Q}(\alpha)^\times$  for a root  $\alpha$  of  $f$ .*

*Proof.* Eliminating  $v_n$  from (2) gives

$$u_{n+2} = (a_n + a_{n+1})u_{n+1} - a_n^2 u_n. \quad (4)$$

Let  $\ell$  be a prime and  $k$  an index with  $\ell \mid u_k$ . Reading (4) at  $n = k-1$  modulo  $\ell$  gives  $u_{k+1} \equiv -a_{k-1}^2 u_{k-1}$ , hence

$$-u_{k-1}u_{k+1} \equiv (a_{k-1}u_{k-1})^2 \pmod{\ell}, \quad (5)$$

so  $-u_{k-1}u_{k+1}$  is a square modulo  $\ell$ .

Let  $\ell \nmid 6m$  and let  $r \in \mathbb{F}_\ell$  be a root of  $f$ . Then  $r \neq 0$  and  $r \neq \pm 1$ , since  $f(0) = \eta \neq 0$  and  $f(\pm 1) = \eta$ . Using  $\eta = r - r^3$ ,

$$f(r-1) = -3r(r-1), \quad f(r+1) = 3r(r+1),$$

so at any index  $k \equiv r-1 \pmod{\ell}$  the polynomial values satisfy  $Q(k) \equiv 0$ ,  $Q(k-1) = -3\kappa r(r-1)$ ,  $Q(k+1) = 3\kappa r(r+1)$  and

$$-Q(k-1)Q(k+1) = 9\kappa^2 r^2 (r^2 - 1). \quad (6)$$

If  $r^2 - 1$  were a nonsquare modulo  $\ell$ , then so would be the right side of (6), because  $9\kappa^2 r^2$  is a nonzero square; agreement at all three of  $k-1, k, k+1$  would then contradict (5). That prohibition recurs at every index of the class  $r-1$  modulo  $\ell$ , and Lemma 2.3 with  $L = 3, h = \ell$  would give  $\underline{d}(S) \geq 1/(3\ell)$ . So  $r^2 - 1$  is a square in  $\mathbb{F}_\ell^\times$  at every root of every good reduction, and Lemma 2.6 applies with  $H(T) = T^2 - 1$ .  $\square$

### 2.3 The cubic field leaves one scale

**Lemma 2.8** (classification of the scales). *Let  $m \geq 1$  and  $c = \pm 1$ , let  $\eta = 6c/m$ , and suppose  $T^3 - T + \eta$  is irreducible over  $\mathbb{Q}$  with a root  $\alpha$  such that  $\alpha^2 - 1$  is a square in  $\mathbb{Q}(\alpha)^\times$ . Then  $m = 12$ .*

*Proof.* Choose  $\beta \in \mathbb{Q}(\alpha)$  with  $\beta^2 = \alpha^2 - 1$  and put  $z = \alpha + \beta$ . Then  $(\alpha + \beta)(\alpha - \beta) = 1$ , so

$$z^{-1} = \alpha - \beta, \quad \alpha = \frac{1}{2}(z + z^{-1}), \quad (7)$$

and  $z$  generates  $\mathbb{Q}(\alpha)$ . Write its minimal polynomial as  $Z^3 + bZ^2 + dZ + w$  with  $b, d, w \in \mathbb{Q}$  and  $w \neq 0$ . From  $T^3 - T + \eta$ ,

$$\text{Tr } \alpha = 0, \quad \text{Tr } \alpha^2 = 2, \quad \text{Tr } \alpha^3 = -3\eta. \quad (8)$$

Newton's identities give  $\text{Tr } z = -b$  and  $\text{Tr } z^{-1} = -d/w$ , so the first equation in (8) and (7) give

$$d = -bw. \quad (9)$$

Next  $\text{Tr } z^2 = b^2 - 2d = b^2 + 2bw$  and  $\text{Tr } z^{-2} = b^2 - 2b/w$ , while  $\text{Tr}(zz^{-1}) = 3$ , so  $4 \text{Tr } a^2 = \text{Tr } z^2 + 6 + \text{Tr } z^{-2}$  and the second equation in (8) reads  $b^2 + b(w - w^{-1}) = 1$ . Multiplying by  $w$  factors this exactly:

$$(bw - 1)(b + w) = 0. \quad (10)$$

Similarly  $\text{Tr } z^3 = -b^3 - 3b^2w - 3w$  and  $\text{Tr } z^{-3} = b^3 - 3b^2/w - 3/w$ , while  $\text{Tr } z + \text{Tr } z^{-1} = 0$  by (9), so  $8 \text{Tr } a^3 = \text{Tr } z^3 + \text{Tr } z^{-3}$  and the third equation in (8) gives

$$\eta = \frac{(b^2 + 1)(w + w^{-1})}{8}. \quad (11)$$

By (10), either  $b = -w$  or  $b = w^{-1}$ , and (11) becomes

$$\eta = \frac{(w^2 + 1)^2}{8w} \quad \text{or} \quad \eta = \frac{(w^2 + 1)^2}{8w^3}.$$

Write  $w = r/s$  with  $r \in \mathbb{Z} \setminus \{0\}$ ,  $s \in \mathbb{Z}_{>0}$  and  $\gcd(r, s) = 1$ . Since  $\eta = 6c/m$ ,

$$m = \frac{48c rs^3}{(r^2 + s^2)^2} \quad \text{or} \quad m = \frac{48c r^3 s}{(r^2 + s^2)^2}. \quad (12)$$

Now  $\gcd(r^2 + s^2, rs) = 1$ , so integrality of  $m$  and  $|c| = 1$  force  $(r^2 + s^2)^2 \mid 48$ , hence  $r^2 + s^2 \in \{1, 2, 4\}$ . With  $r \neq 0$  and  $s \geq 1$  the value 1 is too small, and 4 is not a sum of two nonzero squares, so  $r^2 + s^2 = 2$  and  $|r| = s = 1$ . Then (12) gives  $m = 12cr$ , and  $m > 0$  gives  $m = 12$ .  $\square$

The square condition alone does not exclude the two survivors

$$Q_{12, \pm 1}(n) = 2n(n+1)(n+2) \pm 1. \quad (13)$$

The last step uses the successor coupling that the square condition forgets.

## 2.4 Both survivors fail modulo seven

**Lemma 2.9** (the two forbidden words). *For  $Q_{12,1}$ , every sufficiently late four-index window beginning at  $n \equiv 0 \pmod{7}$  contains an exceptional index. For  $Q_{12,-1}$ , the same holds for windows beginning at  $n \equiv 1 \pmod{7}$ . In either case,  $\underline{d}(S) \geq 1/7$ , and hence also  $\underline{d}(S) \geq 1/28$ . No phase-free four-index obstruction is asserted.*

*Proof.* Reduce modulo 7. For  $c = 1$  the values of  $Q_{12,1}$  at  $n = 0, 1, 2, 3$  are 1, 13, 49, 121, that is

$$(1, 6, 0, 2), \quad (14)$$

and for  $c = -1$  the values at  $n = 1, 2, 3, 4$  are 11, 47, 119, 239, that is

$$(4, 5, 0, 1). \quad (15)$$

Each word recurs in  $n$  with period 7.

Let  $(c_0, c_1, 0, c_3)$  be the residues of  $u$  at four consecutive indices of the primitive tail and  $(d_0, d_1, d_2, d_3)$  those of  $v$ , so that  $c_{i+1} = a_i c_i - d_i$  and  $d_{i+1} = a_i d_i$  modulo 7. The middle zero gives  $a_1 c_1 = d_1$ , then  $d_2 = a_1 d_1$  and  $c_3 = -d_2$ , so

$$d_1^2 = c_1 a_1 d_1 = c_1 d_2 = -c_1 c_3. \quad (16)$$

For (14) this is  $-6 \cdot 2 = 2$ , and for (15) it is  $-5 \cdot 1 = 2$ . The square roots of 2 modulo 7 are 3 and 4, so  $d_1 \in \{3, 4\}$  in both cases.

The first step gives  $a_0 = (c_1 + d_0)/c_0$  and hence  $d_1 = d_0(d_0 + c_1)/c_0$ , which is legitimate because  $c_0 \in \{1, 4\}$  is invertible. For (14) this is  $d_1 = d_0(d_0 + 6)$ , with values

$$(0, 0, 2, 6, 5, 6, 2) \quad \text{at } d_0 = 0, 1, \dots, 6,$$

and for (15) it is  $d_1 = 2d_0(d_0 + 5)$ , with values

$$(0, 5, 0, 6, 2, 2, 6).$$

Both images are  $\{0, 2, 5, 6\}$ , which misses  $\{3, 4\}$ . All seven initial denominator residues have been displayed, so the calculation is exhaustive and rests on no unreported search.

Hence every sufficiently late window  $[7k, 7k + 3]$  for the plus profile, and every sufficiently late window  $[1 + 7k, 4 + 7k]$  for the minus profile, contains an element of  $S$ . Windows within each family are pairwise disjoint. There are  $X/7 + O(1)$  such windows contained in  $[0, X]$ , so choosing one exceptional index in each gives  $\#(S \cap [0, X]) \geq X/7 - O(1)$  and therefore  $\underline{d}(S) \geq 1/7$ . This stronger local bound is not promoted to a uniform lower bound for arbitrary cubic profiles.  $\square$

*Proof of Theorem 2.2.* Assume  $\underline{d}(S) = 0$ . Lemmas 2.4 and 2.5 put the primitive tail in the shape (3) with  $c = \pm 1$  and  $Q_{m,c}$  irreducible. Each excluded branch already contradicts the assumption by producing a positive, branch-dependent lower density, namely  $1/(12\ell)$  or  $1/\ell$  for the relevant prime  $\ell$ . Proposition 2.7 and Lemma 2.8 force  $m = 12$ , and Lemma 2.9 then gives  $\underline{d}(S) \geq 1/28$ .  $\square$

## 2.5 From a regular rate to an exact cubic

**Lemma 2.10** (integer extraction at a regular rate). *Let  $\lambda > 1$  and let  $C_n$  be positive integers with*

$$\frac{C_{n+1}}{C_n} = 1 + \frac{\lambda}{n} + o(n^{-\lambda}). \quad (17)$$

*Then  $\lambda$  is an integer  $d \geq 2$ , and there are  $A \in \mathbb{Q}_{>0}$  and  $B \in \mathbb{Q}$  with  $C_n = A n(n+1) \cdots (n+d-1) + B$  for all large  $n$ .*

*Proof.* Put  $F_n = \Gamma(n + \lambda)/\Gamma(n)$ , so that  $F_{n+1}/F_n = 1 + \lambda/n$  exactly and  $F_n \sim n^\lambda$ . Write  $\varepsilon_n = o(n^{-\lambda})$  for the error in (17) and  $z_n = C_n/F_n$ , so that  $z_{n+1}/z_n = 1 + \varepsilon_n/(1 + \lambda/n)$ . Since  $\lambda > 1$  the errors are absolutely summable, so the product converges and  $z_n \rightarrow K$  for some  $K > 0$ , with  $z_n - K = o(n^{1-\lambda})$  by estimating the tail of the convergent logarithmic product. Hence

$$\delta_n := C_n - KF_n = F_n(z_n - K) = o(n). \quad (18)$$

The recurrence gives  $\delta_{n+1} - \delta_n = (\lambda/n)\delta_n + \varepsilon_n C_n$ , in which the first term is  $o(1)$  by (18) and the second is  $o(1)$  because  $C_n = O(n^\lambda)$ . So  $\Delta\delta_n \rightarrow 0$ , and therefore  $\Delta^j\delta_n \rightarrow 0$  for every  $j \geq 1$ .

For  $F$ , the identity  $\Gamma(n+1+\lambda)/\Gamma(n+1) - \Gamma(n+\lambda)/\Gamma(n) = \lambda \Gamma(n+\lambda)/\Gamma(n+1)$  iterates to

$$\Delta^j F_n = \lambda(\lambda-1) \cdots (\lambda-j+1) \frac{\Gamma(n+\lambda)}{\Gamma(n+j)},$$

whose right side is  $O(n^{\lambda-j})$ . Choose an integer  $j > \lambda$ ; then  $\Delta^j F_n \rightarrow 0$ , so  $\Delta^j C_n \rightarrow 0$ . These are integers, so  $\Delta^j C_n = 0$  for all large  $n$ , and  $C_n$  agrees eventually with a polynomial with rational coefficients. Its growth  $C_n \asymp n^\lambda$  forces the degree to be  $\lambda$ , so  $\lambda = d$  is an integer, and  $d \geq 2$  because  $\lambda > 1$ . For that integer  $F_n = n(n+1) \cdots (n+d-1)$  exactly, so (18) says that the difference of two polynomials of degree  $d$  is  $o(n)$ , hence constant. This is the stated shape with  $A = K > 0$ .  $\square$

The precision in (17) carries weight. The positive integers  $C_n = n(n+1)(n+2) + (-1)^n$  satisfy  $C_{n+1}/C_n = 1 + 3/n + O(n^{-3})$  and are not eventually polynomial, so the exponent cannot be relaxed to  $O(n^{-\lambda})$ . That countermodel is a scalar sequence rather than an exact orbit, and it bears on Lemma 2.10 alone.

**Lemma 2.11** (the tail ratio reads the growth defect). *Let  $a_n$  be strictly increasing positive integers with  $a_{n+1}/a_n^2 \rightarrow 1$  and  $\sum_n 1/a_n$  rational, and let  $(C_n, D_n)$  be the canonical integer tail. Write  $\gamma_n = a_n^2/a_{n+1} - 1$ . Then*

$$\frac{C_{n+1}}{C_n} = 1 + \gamma_n + O(a_n^{-1}), \quad a_n \geq \exp(c 2^n) \text{ eventually for some } c > 0.$$

*Proof.* Since  $a_{n+1}/a_n^2 \rightarrow 1$  and  $a_n \rightarrow \infty$ , there is an  $N$  with  $a_{n+1} \geq a_n^2/2 \geq 2a_n$  for  $n \geq N$ . Iterating  $a_{n+1} \geq a_n^2/2$  from an index where  $a_n > 2$  gives  $a_n \geq \exp(c 2^n)$  eventually. The same doubling gives, for  $m > N$ ,

$$\frac{1}{a_m} \leq x_m = \sum_{k \geq m} \frac{1}{a_k} \leq \frac{2}{a_m},$$

so  $x_m = (1 + \rho_m)/a_m$  with  $0 \leq \rho_m = a_m x_{m+1} \leq 2a_m/a_{m+1} \leq 4/a_m$ , using  $a_{m+1} \geq a_m^2/2$ . Hence

$$\frac{C_{n+1}}{C_n} = \frac{a_n D_n x_{n+1}}{D_n x_n} = a_n \cdot \frac{(1 + \rho_{n+1})/a_{n+1}}{(1 + \rho_n)/a_n} = \frac{a_n^2}{a_{n+1}} \cdot \frac{1 + \rho_{n+1}}{1 + \rho_n}.$$

The last factor is  $1 + O(1/a_n)$  and  $a_n^2/a_{n+1}$  is bounded, so the product is  $a_n^2/a_{n+1} + O(a_n^{-1})$ .  $\square$

Equation (20) of Section 6 records the same comparison in the centred coordinates, where it follows from the kernel-checked defect identity of Theorem 5.1.

*Proof of Theorem 2.1.* The hypothesis  $a_{n+1}/a_n^2 = 1 + 3/n + o(n^{-3})$  gives  $a_{n+1}/a_n^2 \rightarrow 1$ . Suppose  $\sum_n 1/a_n$  were rational and pass to the canonical integer tail, so that  $(a, C, D)$

satisfies (1) with every  $C_n$  and  $D_n$  a positive integer. By Lemma 2.11 and  $a_n \geq \exp(c2^n)$ , the error term  $O(a_n^{-1})$  is  $o(n^{-3})$ , so

$$\frac{C_{n+1}}{C_n} = 1 + \frac{3}{n} + o(n^{-3}).$$

Lemma 2.10 at  $\lambda = 3$  gives  $C_n = A n(n+1)(n+2) + B$  for all large  $n$ , with  $A \in \mathbb{Q}_{>0}$  and  $B \in \mathbb{Q}$ . That contradicts Theorem 2.2.  $\square$

The composition handles every rational normalisation of the tail. It does not assume that the primitive leading coefficient is 1, or 2, or bounded in advance.

**Evidence.** Ordinary proof. The finite claims are the two words (14) and (15), the image  $\{0, 2, 5, 6\}$  of the first-step recursion modulo 7, the divisor condition  $(r^2 + s^2)^2 \mid 48$ , and the countermodel  $n(n+1)(n+2) + (-1)^n$ ; all four are displayed above and replayed exactly. No Lean declaration covers Theorem 2.1 or Theorem 2.2; the centred-coordinate form of Lemma 2.11 rests on the [defect identity](#), which is kernel-checked. Lemma 2.6 uses the Chebotarev density theorem in its finite-Galois form and no effective or unproved input.

**Position.** Theorem 2.1 carries no rationality hypothesis and no state hypothesis, and it is the only unconditional irrationality statement in this note. Its rate class lies inside the growth hypothesis of Problem 1.1, because  $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$  forces  $a_{n+1}/a_n^2 \rightarrow 1$ . On that class the assertion of Problem 1.1 therefore holds, by exclusion of a rational reciprocal sum, and the recurrence is never exhibited; Sylvester’s own sequence, whose ratio  $a_n^2/a_{n+1}$  is  $1 + O(1/a_n)$ , lies outside the class. The rate  $1 + 3/n$  sits strictly beyond the two sufficient hypotheses recorded in Section 3: it fails Koizumi’s  $1 + o(1/n)$  condition, and  $\sum_n \gamma_n$  diverges, so Duverney’s convergent-defect hypothesis does not apply. Theorem 2.2 is independent of any growth or positivity assumption on  $(a_n)$  beyond (1), and the later sections of this note decide nothing at the cubic rate.

### 3 Prior work, and the status of the note

**Two published sufficient conditions.** Two published results reach the original problem under analytic hypotheses that are assumed nowhere in this note. Their hypotheses are conditions on the sequence  $(a_n)$  itself. Duverney’s is not comparable with the state-level hypotheses used here; the Erdős–Straus one, read through the dictionary of the next paragraph, is implied by nonnegativity of  $E$ , the hypothesis of Theorem 8.1, so that criterion already covers the descent of Section 8 (see [7, Cor. 4(1) and Prop. 1(1), pp. 14–15]). Erdős and Straus proved that if  $\lim a_n/a_{n-1}^2 = 1$ , the reciprocal sum is rational, and  $a_n$  does not satisfy the recurrence, then

$$\limsup_{n \rightarrow \infty} \frac{[a_1, \dots, a_n]}{a_{n+1}} \left( \frac{a_{n+1}^2}{a_{n+2}} - 1 \right) > 0,$$

where  $[a_1, \dots, a_n]$  is the least common multiple [1, Theorem 3, p. 132]. This is the indexing in the original theorem. Erdős’s 1988 survey, followed by the current catalogue

summary, instead prints  $a_n^2/a_{n+1} - 1$  in the second factor while retaining the same prefix-LCM quotient; that displayed summary is off by one and is not followed here [5, p. 105][8]. Koizumi records the convenient sufficient rate  $a_n^2/a_{n+1} = 1 + o(1/n)$  under which the Erdős–Straus criterion settles the problem [7, Remark 3, p. 16].

Duverney proved a conditional signed form of the problem itself: for signs  $\epsilon_n \in \{-1, 1\}$ , if  $\sum_{n \geq 0} (a_{n+1}/a_n^2 - 1)$  converges, then the reciprocal sum with numerators  $\epsilon_n$  is rational if and only if

$$a_{n+1} = a_n^2 - (\epsilon_{n+1}/\epsilon_n)a_n + \epsilon_{n+2}/\epsilon_{n+1}$$

for all large  $n$  [2, Corollary 3.2, p. 287]. Absolute convergence is a stronger sufficient specialisation, not the hypothesis printed in that corollary. The all-positive specialisation is the form relevant here.

Badea’s positive-term criterion is adjacent but different: for positive rational terms  $b_n/a_n$ , eventual strict inequality

$$a_{n+1} > \frac{b_{n+1}}{b_n}a_n^2 - \frac{b_{n+1}}{b_n}a_n + 1$$

forces irrationality, while rationality under the corresponding non-strict inequality forces eventual equality [3, Theorem A, p. 313; Cor. 2.2, p. 316]. These positive-term hypotheses do not cover the mixed-sign state orbits studied here. Neither result is formalised here, and the results below do not recover either of them.

Koizumi’s work on doubly exponential sequences [7] bears directly on the state system used below, and supplies the coordinates in which several of the statements of this note are most naturally read. For a rational  $r = p/q$  with pseudo-greedy expansion  $(a_n)$ , remainders  $(x_n)$  and gap sequence  $\epsilon_n = x_n^{-1} + 1 - a_n$ , his Lemma 4 [7, pp. 11–12] produces integers  $c_n > 0$ ,  $d_n$  and  $e_n$  with  $x_n = c_n/d_n$  and  $\epsilon_n = e_n/c_n$ , satisfying

$$a_n = \frac{d_n - e_n}{c_n} + 1, \quad c_{n+1} = c_n - e_n, \quad d_{n+1} = a_n d_n,$$

and the proof of that lemma also gives  $c_{n+1} = a_n c_n - d_n$ . Under the dictionary  $(C_n, D_n, E_n) = (c_n, d_n, e_n)$  these are exactly the objects of Section 4: the first relation is the centring map ctr rewritten as  $e_n = d_n - (a_n - 1)c_n$ , the second is Proposition 4.1, the third is the denominator update, and the fourth is the tail update. The lower-case  $e_n$  of Section 4 is  $-E_n$ , so it is the negative of Koizumi’s  $e_n$  at an index where the centred state is negative.

His Theorem 3 [7, pp. 12–13] shows that his Conjecture 6, that for a positive rational  $r$  whose gap sequence satisfies  $\epsilon_n \rightarrow 0$  one has  $\epsilon_n = 0$  for all large  $n$ , is equivalent to an affirmative answer to the question of Erdős and Graham recorded as his Question 5, which is Problem 1.1 above. Two of the implications proved below have prior art in these canonical coordinates: his Lemma 3, that  $\epsilon_n = 0$  forces  $\epsilon_{n+1} = 0$ , is the absorption of Theorem 5.7, and his Proposition 1(2), that  $\epsilon_n \geq 0$  for all large  $n$  forces  $\epsilon_n = 0$  for all large  $n$ , is the descent of Theorem 8.1. These appear in canonical coordinates in [7, Lemma 3, p. 10; Prop. 1(2), pp. 14–15]. Koizumi attributes Proposition 1(2) to Badea. Its contrapositive says that a counterexample must have negative errors infinitely often;

it assumes the sign is eventually nonnegative, while Theorem 12.1 below assumes only that the negative part is eventually bounded, which admits sign changes of bounded depth at every scale.

The growth hypothesis in Problem 1.1 is calibrated by two facts about Sylvester's sequence. It satisfies  $a_n \approx c_0^{2^n}$  with  $c_0 = 1.2640847\dots$ , and shifting it further produces sequences with  $a_n \approx C^{2^n}$  for arbitrarily large  $C$  whose reciprocals still sum to a rational number [6, arXiv v4, p. 2]. The classical sufficient condition for irrationality,  $\lim_n a_n^{1/2^n} = \infty$ , is therefore sharp. Kovač and Tao identify that condition as folklore [6, arXiv v4, p. 2]; they attribute the sharpness observation to Erdős (1975). A sequence with  $a_n/a_{n-1}^2 \rightarrow 1$  has  $a_n^{1/2^n}$  convergent, so that criterion says nothing about the sequences of Problem 1.1, and rationality is genuinely possible there. Sylvester's sequence is A000058 in the OEIS and its shifts A129871. For the recent literature on irrationality of Ahmes series we refer to Kovač and Tao [6], who resolve several problems of Erdős and Graham drawn from the same two sources cited above, and whose introduction gives a sample of the intermediate work, including Sándor (1984) and Badea (1987). They do not treat Problem #243; the rigidity conclusion asked for there is not among their results.

The *Formal Conjectures* collection contains a mathematically equivalent unproved declaration, up to its zero-based indexing [9]. Its summand is  $\mathbb{Q}$ -valued, so Lean's Summable hypothesis asserts the existence of a sum in  $\mathbb{Q}$ ; the finite indexing shift changes that sum only by a rational prefix. The declaration therefore does encode the rationality premise, but its proof is sorry. Its role here is statement-level prior art; it supplies no proof authority, and the development in this note is independent of it. The machine-checked results concern the state system: they exclude constant and periodic negative magnitudes, the bounded-rise barrier of Theorem 11.3, and the two conditional endpoints (Theorems 12.1 and 13.1), each stated with the exact hypotheses that separate it from Problem #243.

*Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

*Companion system context.* The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.

**Strongest result.** If strictly increasing positive integers satisfy  $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$ , then  $\sum_n 1/a_n$  is irrational (Theorem 2.1). The theorem assumes that rate and nothing else. The rate class sits inside the growth hypothesis of Problem 1.1, so the assertion of the problem holds on it, by exclusion of a rational reciprocal sum; the recurrence is never exhibited. This is an ordinary proof with an exact finite replay, given in full in Section 2, and no Lean declaration covers it.

**Strongest statement under the problem's own hypotheses.** Under the hypotheses of Problem 1.1 and no others, the sequence is eventually Sylvester if and only if the weighted record-excess sum  $\sum_{n \in \mathbb{R}} (-V_n - B)_{+} f(U_n)$  is finite for some integer  $B \geq 0$  (Theorem 6.2). This is an ordinary proof whose finite first-crossing arithmetic is kernel-checked and whose analytic half is not. The hypotheses of Problem 1.1 do not supply that finiteness, so the criterion is a target-equivalent reformulation of the problem rather than a solution of it.

**Strongest Lean-checked result.** Under the exact integer dynamics, positivity, normalised vanishing, and an eventually bounded negative part, the centred error  $E_n$  is eventually zero. The original sequence therefore satisfies the Sylvester recurrence from some point onward (Theorem 12.1), in the bounded-negative-part recurrence theorem (boundedNegativePart\_sylvesterNext\_eventually is not in this check-out). Hypothesis (5), the bound on the negative part, is supplied by nothing in this note.

**Second endpoint.** The same conclusion holds when the normalised negative mass  $\sum_n (-E_n)_{+} / C_n$  is finite, provided the strict centred-step equation is available (Theorem 13.1).

**What remains open.** Koizumi's coordinates provide normalised vanishing for the canonical orbit, but they do not provide either bounded negative excursions or finite normalised negative mass. Any surviving counterexample must therefore have cofinally unbounded negative excursions and divergent normalised negative mass (Proposition 14.1), record increments strictly above the log-log scale (Theorem 7.1), and unbounded record-amplified error (Theorem 7.3).

**Proof spine.** The record criterion charges each first crossing of a CRT-selected height to the record excess at the step that crosses it, and the crossing heights carry a divergent weight. For the state-system endpoints, bounded negative errors control the rise of the integer tail state; a gcd stabilisation reduces the orbit to pairwise-coprime fresh moduli, and a block of consecutive multiples then gives the contradiction. The finite-mass argument replaces the uniform bound by a summable recovery estimate.

The remaining exclusions sharpen this boundary. By Theorem 5.7, a counterexample has  $E_n \neq 0$  for every large  $n$ . By Theorem 8.1 it cannot have  $E$  eventually nonnegative. If its tail is all negative, Theorems 9.2 and 10.1 exclude, respectively, constant magnitude and periodic magnitude in the stated regime  $e_n < a_n$  with positive drift; they are not a classification of a mixed-sign tail. Under normalised vanishing, Theorems 12.1 and 13.1 show that the negative part is neither bounded nor of finite normalised mass. What survives is stated in Section 14.

## Structure

Section 2 proves the unconditional cubic-rate irrationality theorem, which is the only result here that settles the assertion of Problem 1.1 on a class of sequences. Section 4 sets up the state system, and Section 5 proves the defect identity, from which eventual vanishing of  $E$  forces the Sylvester recurrence and a single vanishing error propagates. Section 6 then proves the exact record-excess criterion, and Section 7 states the further criteria in the record coordinates. Section 8 disposes of the case  $E \geq 0$  in three lines.

The rest of the note is the case  $E < 0$ , in four widening steps: constant magnitude (Section 9), periodic magnitude (Section 10), bounded magnitude via a coprimality barrier (Sections 11 and 12), finite normalised negative mass (Section 13), and what is left (Section 14). These sections contain the new content; a reader who wants only the strongest statements should read Section 2, then Sections 6 and 7, and then Section 14. Appendix A is a guide to the formal sources, and Appendix B records an exact residue reduction that an earlier finite search used. Linked phrases open the corresponding Lean declaration at the pinned source revision 92b88dc1bbe0. Where a section names a Lean module that is outside that revision, it says so in place of a link.

## 4 The centred integer state

The three maps defined next carry out the two steps announced at the end of Section 1. The first two clear denominators along the sequence, so that a rational reciprocal sum leaves a sequence of integers; the third subtracts from that integer the value it takes on Sylvester's sequence, leaving one number whose vanishing is the whole question. All three are maps between integer tuples and carry no hypothesis of any kind.

For  $a, D, C \in \mathbb{Z}$  define

$$\text{den}(a, D) = aD, \quad \text{tail}(a, D, C) = aC - D, \quad \text{ctr}(a, D, C) = D - (a - 1)C,$$

the **denominator update**, the **tail update**, and the **centred state**. Given a sequence  $(a_n)$  of integers, whose terms we call the *multipliers*, we write  $D_{n+1} = \text{den}(a_n, D_n)$  for the *denominator state*,  $C_{n+1} = \text{tail}(a_n, D_n, C_n)$  for the *tail state*, and  $E_n = \text{ctr}(a_n, D_n, C_n)$  for the *centred state*, also called the *error*. The word *centred* records what the third map does: it measures  $D_n$  against  $(a_n - 1)C_n$ , and  $D_n = (a_n - 1)C_n$  holds at every index on Sylvester's sequence (Example 4.2), so  $E$  is a deviation from that sequence rather than a size. At an index where  $E_n < 0$  we write  $e_n = -E_n > 0$  and call it the *magnitude* of the error there.

**Proposition 4.1** (update law).  $\text{tail}(a, D, C) = C - \text{ctr}(a, D, C)$ ; that is,  $C_{n+1} = C_n - E_n$ .

*Proof.*  $C - (D - (a - 1)C) = aC - D$ . □

Formalised as the **update law**. The state therefore moves by exactly the error at each step, which is what makes the sign of  $E$  decisive: where  $E$  is positive the state falls, where  $E$  is negative it rises, and where  $E$  vanishes it is stationary.

**Example 4.2** (the Sylvester orbit). Take  $a_n = 2, 3, 7, 43, 1807, \dots$  with  $a_{n+1} = a_n^2 - a_n + 1$ , and start the state at  $D_0 = C_0 = 1$ . The two updates give

| $n$ | $a_n$ | $D_n$ | $C_n$ | $E_n$ |
|-----|-------|-------|-------|-------|
| 0   | 2     | 1     | 1     | 0     |
| 1   | 3     | 2     | 1     | 0     |
| 2   | 7     | 6     | 1     | 0     |
| 3   | 43    | 42    | 1     | 0     |
| 4   | 1807  | 1806  | 1     | 0     |

and  $D_n = a_n - 1$  with  $C_n = 1$  at every index: if  $D_n = a_n - 1$  and  $C_n = 1$  then  $C_{n+1} = a_n - (a_n - 1) = 1$  and  $D_{n+1} = a_n(a_n - 1) = a_{n+1} - 1$ . Hence  $E_n = D_n - (a_n - 1)C_n = 0$  throughout and the tail state never moves, which is Proposition 4.1 in the stationary case.

**The intended reading, which is not formalised.** Let  $T_n = \sum_{k \geq n} 1/a_k$  and suppose  $T_0 \in \mathbb{Q}$ . Put  $D_0$  equal to a common denominator and  $D_n = D_0 a_0 \cdots a_{n-1}$ , and set  $C_n = D_n T_n$ . Then  $C_n \in \mathbb{Z}$  for every  $n$ , and from  $T_n = 1/a_n + T_{n+1}$  one gets  $C_{n+1} = a_n C_n - D_n$ , which is the tail update. On Sylvester's sequence  $T_n = 1/(a_n - 1)$  exactly, so  $D_n = (a_n - 1)C_n$  and  $E_n = 0$ : the centred state measures deviation from the Sylvester tail identity, and it vanishes identically on the Sylvester orbit.

The Lean module carries no rationality hypothesis and no growth hypothesis; it proves identities and implications about the integer system above, and about its natural-number realisation, for which the bridging identities are the [tail realisation](#), the [denominator realisation](#), and the [signed update law](#).

The series itself is reached by three further declarations, which were absent from earlier revisions of this note. Write  $C_n/D_n$  for the real ratio carried by the natural state, the [tail ratio](#). Given  $a_n > 0$ ,  $D_n > 0$ ,  $C_{n+1} + D_n = a_n C_n$  and  $D_{n+1} = a_n D_n$ , one exact step removes  $1/a_n$  from that ratio [one-step reciprocal identity](#), finite iteration telescopes to a partial sum plus the endpoint ratio [finite telescoping](#), and convergence of the endpoint ratio to zero identifies the infinite sum [the series realisation](#):  $\sum_n 1/a_n$  then has sum  $C_0/D_0$ . These three carry no growth hypothesis, no centring hypothesis and no rationality hypothesis, and they exclude nothing. They close the loop between the integer system and the series that motivates it. Producing the state of a given rational sum, which is the direction the problem needs, is the exposition of this paragraph and is not a checked statement.

**Proposition 4.3** (scale equivariance). *For every  $s, a, D, C \in \mathbb{Z}$ ,*

$$\text{den}(a, sD) = s \text{ den}(a, D), \quad \text{tail}(a, sD, sC) = s \text{ tail}(a, D, C),$$

$$\text{ctr}(a, sD, sC) = s \text{ ctr}(a, D, C).$$

Formalised as the [denominator equivariance](#), the [tail equivariance](#), and the [centred equivariance](#). The system depends only on the ray through  $(D, C)$ . This is used twice below: once to normalise the finite search of Appendix B, and once, in Theorem 10.1, as the induction step that removes common scale.

## 5 The defect identity and its two consequences

Section 4 connects the Sylvester recurrence to the error in one direction only: on Sylvester's sequence the error vanishes. The converse is what the problem needs, and it comes from a single identity, which expresses the deviation of  $a_{n+1}$  from the Sylvester successor, multiplied by the tail state  $C_{n+1}$ , as a combination of the errors at  $n$  and at  $n+1$ , the first weighted by  $a_n^2$ . Its two consequences are Theorem 5.6, in which an eventually vanishing error forces the recurrence provided the tail state is eventually nonzero, and

Theorem 5.7, in which, under strict centring (the condition  $|E_n| < C_n$  at every index), one vanishing error propagates to every later index.

Write  $\text{def}(a, a') = a' - \text{syl}(a)$  for the deviation of the next term from the Sylvester successor, the [Sylvester defect](#).

**Theorem 5.1** (defect identity). *For all  $a, a', D, C \in \mathbb{Z}$ ,*

$$\text{def}(a, a') \cdot \text{tail}(a, D, C) = a^2 \text{ctr}(a, D, C) - \text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)),$$

that is,  $\Delta_n C_{n+1} = a_n^2 E_n - E_{n+1}$ .

*Proof.* A direct expansion: both sides equal  $a' a C - a' D - a^3 C + a^2 D + a^2 C - a D - a C + D$ .  $\square$

Formalised as the [defect identity](#). It is a polynomial identity in four indeterminates, with no hypothesis of any kind, and it carries the whole argument twice over: once for rigidity, and once for absorption.

**Example 5.2** (one defect and the error it creates). Continue Example 4.2 but replace  $a_3 = 43$  by  $a_3 = 44$ . The states  $D_3 = 42$  and  $C_3 = 1$  are unchanged, since they depend only on  $a_0, a_1, a_2$ , and  $E_2 = 0$  still. The defect is  $\Delta_2 = a_3 - \text{syl}(a_2) = 44 - 43 = 1$ , and  $E_3 = D_3 - (a_3 - 1)C_3 = 42 - 43 = -1$ , so the identity reads

$$\Delta_2 C_3 = 1 \cdot 1 = 1 = 49 \cdot 0 - (-1) = a_2^2 E_2 - E_3.$$

By Proposition 4.1 the tail state then rises,  $C_4 = C_3 - E_3 = 2$ . A single unit of defect at one index has produced a negative error of magnitude 1 at the next, and the state has started to climb.

**Theorem 5.3** (two vanishing errors force the step). *Let  $a, a', D, C \in \mathbb{Z}$  with  $\text{tail}(a, D, C) \neq 0$ . If  $\text{ctr}(a, D, C) = 0$  and  $\text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)) = 0$ , then  $a' = \text{syl}(a)$ .*

*Proof.* By Theorem 5.1 the product  $\text{def}(a, a') \cdot \text{tail}(a, D, C)$  is zero, and the second factor is nonzero by hypothesis.  $\square$

Formalised as the [local rigidity step](#). The hypothesis  $C_{n+1} \neq 0$  is not removable: at  $C_{n+1} = 0$  the identity gives no information about  $a'$ .

## 5.1 Two-step cancellation and feedback

The one-step state equations hide a second-order constraint once the raw numerator is reduced at two consecutive stages. Write  $u, u_1, u_2$  for the primitive numerator coordinates,  $v, v_1$  for the corresponding denominator coordinates, and  $h, h_1$  for the two common factors removed in reduction. If the two steps have multipliers  $a, a_1$ , then the exact hypotheses are

$$hu_1 + v = au, \quad hv_1 = av, \quad h_1 u_2 + v_1 = a_1 u_1.$$

**Theorem 5.4** (two-step primitive recurrence). *Let  $a, a_1, u, u_1, u_2, v, v_1, h, h_1$  be integers satisfying*

$$hu_1 + v = au, \quad hv_1 = av, \quad h_1 u_2 + v_1 = a_1 u_1.$$

*Then  $a^2 u + h h_1 u_2 = h(a + a_1) u_1$ .*

*Proof.* Multiply the first equation by  $a$ , replace  $av$  using the second equation, and then replace  $h_1u_2 + v_1$  using the third. The remaining terms factor as the displayed right-hand side.  $\square$

This is the [two-step primitive recurrence](#). The hard point is keeping both changing cancellation factors visible. The result is stronger than a one-step congruence, though it remains non-autonomous and cannot by itself force  $a_1 = a^2 - a + 1$ .

There is a complementary integer identity when the relevant step is cancellation-free. Put  $q = ap - p_1$ , and suppose the next primitive coordinate satisfies

$$p_2 + a^2p = (a + a_1)p_1.$$

Then

**Theorem 5.5** (cancellation-free curvature identity). *Let  $a, a_1, p, p_1, p_2, q$  be integers with  $q = ap - p_1$  and  $p_2 + a^2p = (a + a_1)p_1$ . Then*

$$q^2 + (pp_2 - p_1^2) = (a_1 - a)pp_1.$$

The square  $q^2$  is the local remainder, while  $pp_2 - p_1^2$  is a discrete curvature term. The right side records the change in multiplier, so the identity supplies a diagnostic for an anti-shadowing argument without asserting a sign or a global monotonicity. It is checked as [the cancellation-free curvature square](#); no global anti-shadowing conclusion is claimed.

Finally, the reduced denominator transport has a precise realizability gate. Let  $e, e_1$  be centred errors and let  $\Delta$  be the Sylvester defect. If

$$hu_1 = u - e, \quad he_1 = a^2e - \Delta(u - e),$$

then the transport equality

$$h((a_1 - 1)u_1 + e_1) = a((a - 1)u + e)$$

has mismatch which factors through  $u - e$ . Consequently, when  $u - e \neq 0$ ,

$$h((a_1 - 1)u_1 + e_1) = a((a - 1)u + e) \iff a_1 = a^2 - a + 1 + \Delta.$$

This is the exact [feedback transport equivalence](#), obtained from the factored defect at [feedback transport defect](#). The nonzero-raw-numerator hypothesis is essential: if  $u - e = 0$ , the factorised mismatch cannot identify  $a_1$ .

These three identities require orbit-level hypotheses that remain unproved: eventual absence of cancellation, a fixed sign for the curvature, or cofinal passage through the feedback gate. The global mixed-sign and unbounded-negative regimes of Problem #243 remain open.

**Theorem 5.6** (sequence form). *Let  $a, D, C : \mathbb{N} \rightarrow \mathbb{Z}$  satisfy  $D_{n+1} = \text{den}(a_n, D_n)$  and  $C_{n+1} = \text{tail}(a_n, D_n, C_n)$ . If  $E_n = 0$  for all sufficiently large  $n$  and  $C_{n+1} \neq 0$  for all sufficiently large  $n$ , then  $a_{n+1} = a_n^2 - a_n + 1$  for all sufficiently large  $n$ .*

Formalised as the [eventual Sylvester recurrence](#). The proof is routine: take the larger of the two thresholds and apply Theorem 5.3 at each later index. Every later section is directed at the hypothesis of this theorem, that is, at forcing  $E$  to vanish eventually.

The defect identity has a second consequence, which is what makes a single vanishing error worth having.

**Theorem 5.7** (zero is absorbing). *Let  $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$  be an exact natural orbit, so  $C_{n+1} + D_n = a_n C_n$  and  $D_{n+1} = a_n D_n$ , and let  $E_n = \text{ctr}(a_n, D_n, C_n)$ . Suppose the centring is strict,  $|E_n| < C_n$  for every  $n$ . If  $E_n = 0$  then  $E_{n+1} = 0$ .*

*Proof.* Putting  $E_n = 0$  in Theorem 5.1 gives  $\Delta_n C_{n+1} = -E_{n+1}$ , so  $C_{n+1}$  divides  $E_{n+1}$ . A multiple of  $C_{n+1}$  of absolute value smaller than  $C_{n+1}$  is zero, and  $|E_{n+1}| < C_{n+1}$  by hypothesis.  $\square$

Formalised as the [absorption of a vanishing centred state](#), with the companion [contrapositive along a tail](#): if zero is absorbing beyond some index and  $E$  does not vanish eventually, then  $E$  is nowhere zero beyond that index.

Absorption changes the shape of the problem. Either  $E$  hits zero once and stays there, or it is never zero. Sections 9 and 10 address the special all-negative case only. The mixed-sign analysis in Sections 12 and 13 instead separates cofinally negative errors from an eventually positive tail.

## 6 Weighted first crossings of LCM records

This section carries the leading result of the note. It is an exact criterion for Problem 1.1, stated under the hypotheses of that problem and no others, and it is reached by a first-crossing argument rather than by a hypothesis on the negative part. Only steps setting a new LCM numerator record contribute, a fixed LCM-scale baseline may be subtracted, and the remaining mass may be logarithmically discounted.

**Evidence.** The result in this section is an ordinary mathematical proof. Its finite arithmetic and charging step are kernel-checked at the pinned revision: the [freshness of a rise](#), the [covered-wall excess bound](#), the [wall count](#), and the [weighted wall charge](#). The global analytic argument, which is the passage from the per-step charge to divergence of the whole sum, is not formalised. An exact finite replay covers 6,008 seed-modulus cases over 6,000 distinct rational seeds, 26,171 nonterminal transitions and 6,008 terminal states, and every case reaches centred zero within 15 steps; finite data of that kind do not prove universal termination.

**What the record coordinates transfer.** The criterion moves the obligation off the whole orbit. Only steps setting a new LCM numerator record contribute, a fixed LCM-scale baseline may be subtracted, and the remaining mass may be logarithmically discounted, so a counterexample has to sustain divergence along a sparse subsequence. The inherited part of the picture is the eventual regime  $E_n \geq 0$ , which Koizumi's Proposition 1(2) closes, and to which the published sign criteria apply once the shift of Proposition 7.11 is subtracted. What is added here is a criterion valid in the complementary regime as well, stated under the hypotheses of Problem 1.1 and no others. Two propositions

bound what these coordinates reach. Proposition 7.11 exhibits cancellation-free blocks on which every classical hypothesis quantity fails by a fixed number of units, and Proposition 7.12 shows that the data used by Theorem 11.3 and the landing lemmas admit walks with rises of order  $\ell(u)$ , so an argument using only those data excludes no negative part of that order. A producer that closes the gap has to enter the regime displayed in Section 7, where the negative part is at least  $(1 - \delta) \log_2 \log_2 C_n$  and is  $o(C_n)$  at infinitely many indices, and inside that regime it has to use the congruence  $E_n \equiv D_n \pmod{C_n}$  beyond coprimality, or supply a second mechanism that forces landings.

**Remaining obligation.** Derive finiteness of the displayed sum for one admissible weight and one fixed  $B$  from rational denominator feedback, or contradict a nonterminal orbit by another route. The logarithmic discount weakens what suffices; it does not establish it.

Set

$$L_n = \text{lcm}(q, a_0, \dots, a_{n-1}), \quad M_n = D_n/L_n, \quad U_n = C_n/M_n, \quad V_n = E_n/M_n.$$

The rational tail has denominator dividing  $L_n$ , so  $U_n$  and  $V_n$  are integers. With  $\rho_n = \gcd(L_n, a_n)$ , the exact updates are

$$M_{n+1} = M_n \rho_n, \quad \rho_n U_{n+1} = U_n - V_n, \quad V_n = L_n - (a_n - 1)U_n.$$

Write  $R_n = \max_{j \leq n} U_j$  and  $\mathcal{R} = \{n : U_{n+1} > R_n\}$ . Centring implies that every sufficiently late strict rise has  $\rho_n = 1$ : otherwise  $U_{n+1} \leq 3U_n/4$ . Its actual jump is therefore  $d_n = U_{n+1} - U_n = -V_n$ .

**Theorem 6.1** (arithmetic weighted-record dichotomy). *Let  $a_n, L_n, U_n$  be positive integers and  $V_n$  integers satisfying*

$$\begin{aligned} L_{n+1} &= \text{lcm}(L_n, a_n), & \rho_n &= \gcd(L_n, a_n), \\ \rho_n U_{n+1} &= U_n - V_n, & V_n &= L_n - (a_n - 1)U_n, & -U_n &\leq 2V_n. \end{aligned}$$

*Let  $f : [1, \infty) \rightarrow [0, \infty)$  be finite and nonincreasing, with  $\int_1^\infty f(t) dt = \infty$ . For each fixed integer  $B \geq 0$ ,*

$$\sup_n U_n < \infty \iff \sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

*Proof.* A bounded integer running maximum increases only finitely many times, so bounded  $U_n$  gives a finite sum. Suppose instead that  $U_n$  is unbounded. There are infinitely many record rises, each with  $\rho_n = 1$ . Its digit is greater than one because  $d_n = (a_n - 1)U_n - L_n > 0$ . If  $r < s$  are record indices, then  $a_r \mid L_s$  and  $\gcd(a_s, L_s) = 1$ , hence  $\gcd(a_r, a_s) = 1$ . The record digits are therefore distinct and pairwise coprime. For any fixed  $B \geq 1$ , choose  $B$  of them greater than  $B$ , and take  $T$  after their indices. These old digits  $m_0, \dots, m_{B-1}$  all divide  $L_T$ .

Put  $P = \prod_i m_i$  and choose  $x$  by the Chinese remainder theorem with  $m_i \mid x + i$ . Consider the heights  $\tau = x + B + kP$ . A first crossing  $U_n \leq R_n < \tau \leq U_n + d_n$  is a record step. If  $d_n \leq B$ , then  $U_n \in [\tau - B, \tau)$ , so some  $m_i$  divides  $U_n$ . It also divides  $L_n$ , hence divides  $d_n = (a_n - 1)U_n - L_n$ , contradicting  $0 < d_n \leq B < m_i$ .

If the step first crosses  $h \geq 1$  such heights, their spacing gives  $(h-1)P < d_n$ . With  $r = d_n - B \geq 1$  and  $P \geq B+1$ , we have  $d_n = B + r \leq Pr$ , whence  $h \leq r$ . Monotonicity of  $f$  now gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \leq (d_n - B)f(U_n).$$

Each height above  $R_T$  has one first crossing. Summing, and comparing each interval of length  $P$  with its left endpoint, yields the finite bound

$$\sum_{\substack{T \leq n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \geq \frac{1}{P} \int_{R_T+P}^{R_N} f(t) dt. \quad (19)$$

Since  $R_n \rightarrow \infty$ , the right side diverges for every  $B \geq 1$ . The case  $B = 0$  follows by domination.  $\square$

Only the lower centring bound was used. In particular, neither normalised vanishing nor a growth estimate for  $C_n$  supplies the CRT moduli: the record digits themselves do so. Normalised vanishing enters in the following application to the original sequence.

**Theorem 6.2** (weighted record excess). *Assume the growth and rationality hypotheses of Problem 1.1, and let  $f$  be as in Theorem 6.1. The sequence is eventually Sylvester if and only if, for some integer  $B \geq 0$ ,*

$$\sum_{n \in \mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

*Proof.* If the endpoint fails, centred zero is never reached on a sufficiently late tail. Integrality gives  $1/U_n \leq |V_n|/U_n = |E_n|/C_n \rightarrow 0$ , so  $U_n$  is unbounded. Theorem 6.1, applied after the centring threshold, forces divergence for every  $B$ . Removing a finite prefix changes only finitely many global record terms. Conversely, a Sylvester tail telescopes to  $x_n = 1/(a_n - 1)$ , so  $V_n = 0$  eventually.  $\square$

For example,  $f(t) = 1/[t \log(et)]$  gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (19) is  $P^{-1} \log(\log(eR_N)/\log(e(R_T + P)))$ . Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. These are specialisations of one crossing theorem.

The criterion also has an exact expression in the original growth defect. Put  $\gamma_n = a_n^2/a_{n+1} - 1$  and  $\theta_n = E_n/C_n$ . The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \quad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Thus the two nonnegative summands  $U_n f(U_n)(\gamma_n - B/U_n)_+$  and  $(-V_n - B)_+ f(U_n)$  differ by at most  $3U_n f(U_n)/a_n$ . This is summable because  $U_n \leq C_n =$

$\exp(o(n)), f(U_n) \leq f(1)$ , and  $a_n \geq \exp(c2^n)$  eventually for some  $c > 0$ . Consequently Theorem 6.2 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_n f(U_n) \left( \frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \quad (20)$$

for some  $B$ . The original hypotheses do not currently supply this finiteness. In particular, termwise convergence to zero is insufficient. The same equivalence holds directly on the canonical LCM orbit under the original positive, strictly increasing integer sequence, rational reciprocal-sum and near-square-growth hypotheses. Its summand is exactly  $U_n f(U_n) \max(a_n^2/a_{n+1} - 1 - B/U_n, 0)$  at record indices and zero elsewhere. For every nonnegative nonincreasing  $f$  on  $[1, \infty)$  with divergent integral, summability for some  $B \in \mathbb{N}$  is equivalent to eventual Sylvester recurrence. The checked comparison uses the sufficient constant 16 in place of the sharper ordinary constant 3 above. It proves  $\sum_n C_n/a_n < \infty$  from the canonical tail formula and the ratio test for  $P_n/a_n^2$ , whose consecutive-term ratio tends to zero ( $P_n = \prod_{j < n} a_j$ ). Since  $U_n \leq C_n$  and  $f(U_n) \leq f(1)$ , the comparison error is summable. This transfers the criterion; it does not supply the growth budget from the original hypotheses. The [factored-growth equivalence](#) is checked in Lean under these same hypotheses.

Also,  $d_n$  is the actual jump, including any recovery from a drawdown; it must not be replaced by  $R_{n+1} - R_n$  in the crossing proof.

## 7 Criteria in the record coordinates

The criterion of Section 6 lives in the LCM coordinates  $U_n = C_n/M_n$ , where old divisors persist. The statements of this section live in the primitive coordinates, where old divisors can be deleted. They are collected here because each is an equivalence, a dichotomy, or an unconditional exclusion, and because each sharpens the profile of a counterexample in a direction the earlier sections do not reach.

**Primitive coordinates.** Put  $G_n = \gcd(C_n, D_n)$ ,  $u_n = C_n/G_n$ ,  $v_n = D_n/G_n$ ,  $\tilde{e}_n = E_n/G_n$  and  $h_n = G_{n+1}/G_n$ , so that  $w_n = a_n u_n - v_n = u_n - \tilde{e}_n$ ,  $h_n u_{n+1} = w_n$ ,  $h_n v_{n+1} = a_n v_n$  and  $\gcd(u_n, u_{n+1}) = 1$ . The primitive error  $\tilde{e}_n$  is signed, and  $e_n = -E_n$  of Section 4 remains the magnitude in the unreduced coordinates; neither is Koizumi's rational gap  $\varepsilon_n$  of Section 3. Write  $R_n = \max_{k \leq n} u_k$  and  $H_n = \max_{j \leq n} C_j$  for the two running maxima,  $s_n = R_{n+1} - R_n$  for the true record increment,  $d_n = u_{n+1} - u_n$  at a strict rise,  $m_n = (-\tilde{e}_n)_+$ ,  $A_n = R_n m_n / u_n$  for the record-amplified negative error,  $\delta_n = (a_n^2/a_{n+1} - 1)_+$  for the growth defect, and  $\ell(x) = \log_2 \log_2 \max(4, x)$ . The symbol  $A_n$  is unrelated to the prefix product  $A_n$  of Section 14. A step is *clean* when  $h_n = 1$ . Call the standing hypotheses of Problem 1.1, read through Koizumi's coordinates together with normalised vanishing, the hypotheses (H), and write (S) for the conclusion that the sequence is eventually Sylvester.

**Source and evidence conventions for this section.** Each statement below carries its own evidence line. "Lean" means that a declaration with that statement and a complete proof body is present in the formal sources. Several of the modules named below are

outside the pinned public revision 92b88dc1bbe0; those are marked, and no link is offered for them. Where a result is an ordinary proof over a Lean core, the Lean part is named exactly, and the part that is ordinary mathematics is named exactly as well.

## 7.1 The record-increment coefficient

**Theorem 7.1** (record-excess dichotomy). *Let  $\Theta = \limsup_n (H_{n+1} - H_n)/\ell(H_n)$  on a non-terminating canonical orbit under  $(H)$ . Then either  $G_n$  is unbounded and  $\Theta$  is infinite, or  $G_n$  stabilises at a value  $g$  and  $\Theta \geq g v_T/\varphi(v_T)$  for every late  $T$ , so that  $\Theta > g \geq 1$ . Hence  $\Theta = 0$  or  $\Theta > 1$ , and  $\Theta \leq 1$  forces  $(S)$ .*

**Corollary 7.2** (inclusive log-log boundary). *If  $\limsup_n (-E_n)_+/\ell(C_n) \leq 1$  then  $(S)$  holds. Every counterexample therefore satisfies  $\limsup_n (-E_n)_+/\ell(C_n) > 1$  strictly.*

**Mechanism.** Four steps. The drawdown ladder  $r_n = (-E_n - B_n)_+$  with  $B_{n+1} = (B_n + E_n)_+$  and  $B_n = H_n - C_n$  gives  $0 \leq r_n \leq (-E_n)_+$ , so a bound on the record increment is weaker than a bound on the negative error and is therefore the better trigger. A divisor of  $C_t$  and  $D_t$  at a record locks into every later record increment, so a pairwise-coprime block of moduli above  $B$  dividing  $D_T$  contradicts  $r_n \leq B$  below height  $2P + B$ . If  $\sup_n G_n$  is infinite then  $\Theta$  is infinite, so a finite  $\Theta$  forces a genuinely primitive tail on which all multipliers are pairwise coprime. Pre-sieving the CRT block with  $x \equiv 0 \pmod{v_N}$  makes a block of length  $B$  cost only  $\varphi(v_N)v_N^{-1}B$  fresh moduli, which gives the strict inequality.

**Evidence.** Ordinary proof over the Lean core. The kernel-checked ingredients are the [persistence of a common divisor](#), the [bounded CRT block](#), the [slow-rise landing lemma](#), and the [coprime-block exclusion](#). The running-maximum persistence clause, the record-increment trigger, the gcd rescaling and the pre-sieved amplification are ordinary mathematics. Finite replay: the identities hold exactly on four orbits, the  $B = 20, R = 6$  CRT reconstruction equals 521026757646 and excludes all twenty positions, and the Fermat product identity and the maximal gap ratio 1.8116 at  $u \geq 1000$  against  $\sigma^{-1} = 2$  hold.

**Remaining obligation.** No upper bound on  $\Theta$  is proved. The branch  $\Theta = \infty$  is exactly the surviving regime.

## 7.2 Two exact equivalences under arbitrary cancellation

**Theorem 7.3** (record-amplified rigidity). *Under  $(H)$ , the following are equivalent:  $(S)$ ;  $\limsup_n A_n < \infty$ ;  $\limsup_n R_n \delta_n < \infty$ . Each of those two limits superior is 0 or  $+\infty$ .*

**Corollary 7.4** (the critical rate). *Under  $(H)$  and  $\delta_n = O(1/n)$ , with no convergence of  $n\delta_n$  assumed,  $(S)$  is equivalent to  $u_n = O(n)$  and to  $(-\tilde{e}_n)_+ = O(1)$ . A counterexample at the critical rate therefore has  $\limsup_n u_n/n = \infty$  and  $\limsup_n (-\tilde{e}_n)_+ = \infty$ .*

**Mechanism.** On a non-Sylvester orbit, arbitrarily late indices carry  $B$  distinct primes above a chosen bound dividing  $v_T$ , because paid steps have density zero and a block of consecutive clean steps has pairwise-coprime multipliers all dividing the final denominator. A finite alternative between a CRT crossing and a deletion then forces some  $A_n$  above any prescribed level. The passage to the growth defect uses  $|\delta_n - m_n/u_n| \leq 3/a_n$  and  $R_n/a_n \rightarrow 0$ .

**Evidence.** Ordinary proof. The step that a cancellation at  $s$  is still visible at the first later negative error, in the form  $u_t \leq u_{s+1}$  and  $u_s u_t \leq R_t |\tilde{e}_t| u_{s+1}$ , is Lean-checked as `recordAmplified_error_after_cancellation` in `SaturatedSquareTransport.lean`, which is outside the pinned public revision 92b88dc1bbe0. Exact fixture:  $15/134 = 1/10 + 1/85 + 1/5695$ , with primitive steps  $(15, 134) \rightarrow (4, 335) \rightarrow (1, 5695)$  and  $A_1 = 15/4$ , meeting the visibility lemma with equality.

**Remaining obligation.** The equivalence relocates the problem into the  $A_n$  axis. Nothing here bounds  $A_n$  on the canonical orbit.

**Theorem 7.5** (unit record increments). *Under (H), if  $R_{n+1} - R_n \leq 1$  for all large  $n$ , then  $a_{n+1} = a_n^2 - a_n + 1$  for all large  $n$ . Hence (S) holds if and only if  $\#\{n : R_{n+1} - R_n \geq 2\}$  is finite. No hypothesis is placed on drawdowns, on record-setting jumps, or on the cancellation factors  $h_n$ .*

**Mechanism.** Otherwise  $u_n \rightarrow \infty$ . A late fresh index  $j$  cannot have all its exact prime powers below  $B = R_{j+1} + 2$ , since  $a_j \mid \text{lcm}(1, \dots, B)$  would give  $\log a_j \leq B \log B$  against the doubly exponential growth of  $a_j$ . So some exact prime power  $P = p^k$  of  $a_j$  exceeds  $R_s + 2$  at  $s = j + 1$  and divides  $v_s$ . Let  $Y$  be the least multiple of  $p$  above  $R_s$ . At the first  $t > s$  with  $u_t \geq Y$  the unit increment forces  $u_t = Y$  exactly, while the valuation threshold keeps  $P \mid v_t$ , so  $p \mid u_t$  contradicts primitivity.

**Evidence.** Lean, with an explicit supply hypothesis. The declarations are `unit_recordIncrement_cut` and `recordIncrementOne_sylvesterNext_eventually` in `RecordIncrementBarrier.lean`, which is outside the pinned public revision 92b88dc1bbe0. The prime-power supply is a hypothesis of the formal statement and is discharged at every late index by an ordinary proof, so the complete advertised endpoint is an ordinary proof with a kernel-checked principal part, namely the landing contradiction.

**Relation.** This theorem is incomparable with the two-unit record theorem: a record-setting jump of size at most two bounds  $s_n \leq 2$  rather than  $s_n \leq 1$ , and  $s_n \leq 1$  allows arbitrarily large record-setting jumps after a drawdown. The fixture  $(8, 177) \rightarrow (7, 4071) \rightarrow (10, 2373393)$  with multipliers 23 and 583 has records 8 and 10, increment 2, jump 3 and  $\gcd(8, 10) = 2$ , which is why the parity step of the odd cut does not transfer from jumps to increments.

### 7.3 Protected epochs and the energy criterion

**Theorem 7.6** (protected epoch inequality). *Let the orbit satisfy the primitive cocycle with  $2|\tilde{e}_n| < u_n$  from an index  $s$ . Let  $p \geq 3$  be prime,  $Q = p^\ell$  divide  $v_s$  with  $Q \geq 16$ , put  $L = pQ/2$ , and assume  $R_s < L/2$ . Let  $\tau$  be the first index with  $u_\tau \geq L$ , let  $J$  be the set of steps in  $[s, \tau)$  that first cross at least one odd multiple of  $p$  in  $(L/2, L]$ , and put  $X = \sum_{n \in J} (d_n - 2)$ . Then every  $n \in J$  is a clean record step with  $d_n \geq 3$ , and  $pQ \leq (8p + 8)|J| + 4X + 8p$ .*

**Theorem 7.7** (energy criterion). *Under (H), the energy  $\mathcal{E} = \sum_{n \text{ record}} (\mathbf{1}_{d_n \geq 3} u_n^{-1/2} + (d_n - 2)_+ u_n^{-1})$  is finite if and only if (S) holds; and  $\sum_{n \text{ record}} (d_n - 2)_+ u_n^{-1/2}$  is finite if and only if (S) holds.*

**Mechanism.** Below  $L$  the numerator satisfies  $w_n < \frac{3}{2}L < p^{\ell+1}$ , so the prime power persists and  $p \nmid u_t$  throughout  $[s, \tau]$ . Each barrier above  $R_s$  is first crossed at a record step, where a jump of at most two would require either the forbidden landing or a pair of even endpoints. Barriers first crossed at one step lie in an interval of length  $d_n$  with spacing  $2p$ , which gives the per-step capacity  $1 + d_n/(2p)$ , and the barrier count is at least  $Q/8 - 1$ . For the criterion, an odd prime power of the required size is available at every late index, so each late window carries energy at least  $1/16$ , and the tail of a convergent series tends to zero.

**Evidence.** The integer epoch inequality is Lean, as `protected_epoch_energy_integer` with its three counting lemmas in `ProtectedEpochEnergy.lean`, which is outside the pinned public revision 92b88dc1bbe0. The real-valued corollary, the odd prime-power supply at every late index, and the global criterion are ordinary proofs. This discharges a quantitative record-excess target that an earlier wave had left open, and it uses one protected prime power rather than a product of moduli.

## 7.4 A slow negative part, and the original coordinates

**Theorem 7.8** (slow negative part). *Let  $(a, C, D)$  be an exact natural orbit with  $a_n > 1$ ,  $C_n > 0$ ,  $D_0 \geq 1$ , under normalised vanishing. Suppose that for some  $\delta \in (0, 1)$  and all large  $n$  with  $E_n < 0$  one has  $-E_n \leq (1 - \delta)\ell(C_n)$ . Then  $E_n = 0$  for all large  $n$ , and  $a_{n+1} = a_n^2 - a_n + 1$  for all large  $n$ .*

Theorem 12.1 is the case of a constant bound, since a constant is eventually below  $(1 - \delta)\ell(C_n)$ .

**Mechanism.** Normalised vanishing gives  $C_{n+1} < \frac{3}{2}C_n$ , hence  $\ell(C_n) \leq \log_2 n + O(1)$ , and  $a_n$  grows at least geometrically. The multiplier overlap  $\gcd(a_n, D_n)$  is bounded by the next negative magnitude and therefore by  $\ell(C_n) + 1$ . Splitting  $a_n$  into the part coprime to  $D_n$  and the burnt part, a poor index supplies a burn prime with valuation above  $\sqrt{n}$ , and such primes are injective across a window, so a window  $[T, 2^T]$  has at most  $T + O(1)$  poor indices. The first  $B$  rich indices give  $B$  pairwise-coprime moduli above any prescribed size, and the block form of the landing lemma applies.

**Evidence.** Ordinary proof over the Lean core. The kernel-checked ingredients are the [persistence of a common divisor](#), the [overlap transport](#), the [landing lemma](#) and the [coprime-block exclusion](#). The burn count and the choice of constants are ordinary mathematics. Corollary 7.2 supersedes this theorem on two axes, the drawdown-corrected observable and the inclusive constant, and this theorem is retained because its hypothesis is stated directly on  $E_n$ .

**Theorem 7.9** (the Erdős–Straus quantity, bounded form). *Let  $a_{n+1}/a_n^2 \rightarrow 1$  and  $\sum 1/a_n \in \mathbb{Q}$ , and put*

$$Q_n = \frac{a_1 a_2 \cdots a_{n-1}}{a_n} \left( \frac{a_n^2}{a_{n+1}} - 1 \right).$$

*If  $\limsup_n Q_n < \infty$  then the sequence is eventually Sylvester. If for some  $\delta > 0$  and all large  $n$  one has  $Q_n \leq (1 - \delta)\lambda \ell(a_1 \cdots a_{n-1}/a_n)$ , where  $\lambda > 0$  depends only on the sequence, the same conclusion holds.*

**Attribution.** Erdős and Straus require  $\limsup_n Q_n \leq 0$  in the corresponding criterion, with the least common multiple in place of the product [1, Theorem 3, p. 132]; Koizumi's Corollary 4(1) uses the product form [7, Cor. 4(1), pp. 14–15]. The statement above is the same Lean-checked endpoint as Theorem 12.1, transported into that language, and its content over the published criteria is the passage from  $\leq 0$  to  $< \infty$ . It is the only place in this note where the machine-checked theorem is written in the vocabulary of the 1964 paper.

**Evidence.** Ordinary proof composing Theorem 12.1 and Theorem 7.8 with Koizumi's bridge. The identity used is  $Q_n = -(B_{n-1}/c_1)\varepsilon_n + (B_{n-1}/c_1)\beta_n c_{n+1}$  with  $\beta_k = O(1/a_k)$ , whose second term tends to zero because  $c_{n+1} \leq c_1(3/2)^n$  while  $a_n$  is doubly exponential.

**Theorem 7.10** (the  $1/n$  threshold). *Let  $a_{n+1}/a_n^2 \rightarrow 1$  and  $\sum 1/a_n \in \mathbb{Q}$ . If  $\limsup_n n(a_n^2/a_{n+1} - 1)_+ < 1$ , then  $a_{n+1} = a_n^2 - a_n + 1$  for all large  $n$ .*

The constant 1 is the boundary of this method. The template  $\tilde{e}_n \equiv -1$  with  $u_n = c + n$  has  $n(-\tilde{e}_n)/u_n \rightarrow 1$ , meets every hypothesis of the Erdős–Straus criterion except the sign, and is excluded only by Theorem 9.2. This sharpens the sufficient rate  $a_n^2/a_{n+1} = 1 + o(1/n)$  recorded by Koizumi [7, Remark 3, p. 16] to an explicit constant. Ordinary proof; the endgame is Koizumi's Proposition 1(2).

## 7.5 What the method cannot reach

**Proposition 7.11** (the classical quantities carry a shifted sign). *The overlap debt divides the tail gcd, so  $\kappa'_n = G_n/M_n$  is a positive integer and  $E_n/M_n = \kappa'_n \tilde{e}_n$  is an integer carrying the sign of  $E_n$ . The growth defect carries a shifted sign. On any exact orbit,*

$$\frac{a_n^2}{a_{n+1}} - 1 = -\frac{E_n}{C_n} + \Lambda_n, \quad \Lambda_n = \frac{(1 - E_n/C_n)(a_n - 1 + E_{n+1}/C_{n+1})}{a_{n+1}}, \quad (21)$$

and  $0 < \Lambda_n < 3/a_n$  for all large  $n$  whenever  $E_n/C_n \rightarrow 0$ . The Erdős–Straus quantity of Theorem 3 is  $Q_n = (L_n/a_{n+1})(a_{n+1}^2/a_{n+2} - 1)$ , a positive multiple of the defect at the next index. Thus  $Q_n$  is positive, zero, or negative according as  $E_{n+1}/C_{n+1}$  is less than, equal to, or greater than  $\Lambda_{n+1}$ . In particular, whenever  $\Lambda_{n+1} > 0$ , it is positive if  $E_{n+1} \leq 0$ , and for  $E_{n+1} > 0$  it is negative precisely when  $E_{n+1}/C_{n+1} > \Lambda_{n+1}$ . It is strictly positive at every index of Sylvester's own sequence, where  $E_n = 0$  and  $\Lambda_n = (a_n - 1)/a_{n+1}$ .

*Proof.* The first sentence is the divisibility dictionary of Section 7. For (21), write  $\theta_n = E_n/C_n$ ; Proposition 4.1 gives  $C_{n+1} = C_n(1 - \theta_n)$  and  $D_n/C_n = a_n - 1 + \theta_n$ , so  $a_{n+1} - 1 + \theta_{n+1} = a_n D_n/C_{n+1} = a_n(a_n - 1 + \theta_n)/(1 - \theta_n)$ . Multiplying the asserted identity by  $a_{n+1}$  and substituting reduces it to  $a_{n+1} + a_n - 1 + \theta_{n+1} = a_n^2/(1 - \theta_n)$ , which is the displayed relation with  $a_n$  added to both sides. The bound follows from  $\theta_n \rightarrow 0$  and  $a_{n+1} \geq a_n^2/2$ . On Sylvester's sequence  $\theta_n = 0$  and  $a_{n+1} = a_n^2 - a_n + 1$ , so  $a_n^2/a_{n+1} - 1 = (a_n - 1)/a_{n+1} = \Lambda_n > 0$ .  $\square$

**Reading.** The classical criteria therefore do not decide the half-space  $E_n \geq 0$ . They read the defect  $-E_n/C_n + \Lambda_n$ , whose positive shift  $\Lambda_n$  is of order  $1/a_n$  and is the whole quantity

at the Sylvester locus. A criterion phrased as a sign condition on  $Q_n$  is a condition on the state only at the indices where the negative part dominates the shift. Ordinary proof over exact identities, with an exact finite replay on 4,472 frames and 610 clean triples, zero failures. The fixture family with  $C_n = c + nm$ ,  $D_n = (a_n - 1)C_n - m$  and  $a_n = 1 + C_{n+1}k_n$  realises, at  $(c, m, h) = (10, 3, 6)$ ,  $(13, 3, 6)$  and  $(100003, 3, 5)$ , arbitrarily long blocks on which every classical hypothesis quantity equals  $-E_n = 3$  at every index while  $G_n = \kappa'_n = M_n = 1$ . A third classical quantity, from Tijdeman and Yuan's 2002 Theorem 4.1, enters the same computation; it is not cited here because this note carries no verified bibliographic record of that paper.

**Proposition 7.12** (what whole-modulus avoidance permits). *Let  $m_0 < m_1 < \dots$  be pairwise coprime integers at least 2 with  $\theta = \sum_i 1/m_i < 1$ . For every  $x \geq 1$  and every  $L > k/(1-\theta)$ , where  $k = \#\{i : m_i \leq x + L\}$ , the window  $[x, x + L)$  contains an integer divisible by no  $m_i$ . Consequently, if  $\ell(m_i) = i + O(1)$  then for every  $\epsilon > 0$  some subfamily carries a walk  $u_n \rightarrow \infty$  divisible by no  $m_i$  with  $u_{n+1} - u_n \leq (1 + \epsilon)\ell(u_n)$  for all large  $n$ .*

*Proof.* An integer of  $[x, x + L)$  lies in  $[1, \infty)$  and is smaller than  $x + L$ , so a modulus exceeding  $x + L$  divides none of them. Each of the  $k$  remaining moduli divides at most  $L/m_i + 1$  of them, so the covered count is at most  $L\theta + k < L$ . For the walk, discard an initial segment of the family so that  $\theta < \epsilon/(1 + \epsilon)$ ; then  $k(y) \leq \ell(y) + O(1)$  and  $L = \lfloor k/(1 - \theta) \rfloor + 1 \leq (1 + \epsilon)\ell(y)$  for large  $y$ . Define  $u_{n+1}$  to be an admissible integer of  $[u_n + 1, u_n + 1 + L(u_n))$ , which the window statement supplies.  $\square$

**Reading.** Theorem 11.3 and the landing lemmas use about the multipliers only that they are pairwise coprime, that they divide later denominator states, and that a modulus dividing the current numerator is impossible at a landing, which is whole-modulus avoidance. Proposition 7.12 shows that those data admit walks whose rises reach  $(1 + \epsilon)\ell(u_n)$ , so the bounded-rise hypothesis of Theorem 11.3 cannot be relaxed to a rise of order  $\ell(u)$  by any argument using only them, and the constant 1 in Corollary 7.2 is the CRT constant rather than a feature of the problem. Coprimality to  $m_i$  is a strictly stronger requirement than avoiding its multiples, and the window statement is false for it: the pairwise coprime family  $30, 31, 931, \dots$  has  $\sum_i 1/m_i = 1/15$ , and no integer of  $[2, 5)$  is coprime to 30. Ordinary proof, with a numerical witness walk whose maximal rise against Sylvester's numbers as moduli is 12 up to height  $2 \cdot 10^6$ , and whose maximal ratio  $(u_{n+1} - u_n)/\ell(u_n)$  against the Fermat moduli  $2^{2^j} + 1$  is 1.8116 for  $u \geq 1000$  up to  $3 \cdot 10^6$ . The Fermat product  $\prod_{j \geq 0} (1 - 1/(2^{2^j} + 1))$  telescopes to  $1/2$  exactly; whether that constant is the exact maximal-gap coefficient is Problem 14.7.

**The exact remaining obstacle.** Every criterion of this section leaves the same regime, which is

$$(1 - \delta)\ell(C_n) \leq -E_n = o(C_n) \quad \text{at infinitely many } n,$$

with divergent normalised negative mass, unbounded record-amplified error, and infinitely many clean primitive record jumps of size at least three. Any argument entering it must use the congruence  $E_n \equiv D_n \pmod{C_n}$  beyond coprimality, or supply a second mechanism that forces landings.

## 8 Descent when the error is nonnegative

The hypothesis of the theorem below is the update law of Proposition 4.1 read inside  $\mathbb{N}$ : the error is nonnegative at every index, so the tail state never rises. This is the easy case, and it is settled in three lines.

**Theorem 8.1** (descent). *Let  $C, E : \mathbb{N} \rightarrow \mathbb{N}$  satisfy  $C_{n+1} + E_n = C_n$  for every  $n$ . Then  $E_n = 0$  for all sufficiently large  $n$ .*

*Proof.* A one-line descent. The relation forces  $C_{n+1} \leq C_n$ , so  $C$  is a nonincreasing sequence of natural numbers and is eventually equal to its minimum; beyond that index  $C_{n+1} = C_n$ , and the relation gives  $E_n = 0$ .  $\square$

Formalised as the [stabilisation of the nonnegative error](#), on the [eventual constancy of a nonincreasing sequence](#).

The hypothesis is exactly one-sidedness:  $C$  and  $E$  take values in  $\mathbb{N}$ . If  $E_n < 0$  at infinitely many indices then  $C$  rises there, nothing descends, and the argument is unavailable. That is the whole difficulty, and the rest of this note is directed at it.

## 9 Excluding a constant negative magnitude

Suppose the error is negative with a constant magnitude,  $E_n = -m$  for a fixed  $m > 0$  and every  $n$ . By Proposition 4.1 the state then climbs by exactly  $m$  each step, so  $C_n = c + nm$  with  $c = C_0$ , and the definition of the centred state becomes a *shape equation*

$$D_n + m = (a_n - 1)(c + nm). \quad (5.1)$$

Together with  $D_{n+1} = a_n D_n$  this is a closed system in  $(a, D)$ , and it has no solutions. The following instance shows how the failure happens; the theorem then shows that it cannot be avoided.

**Example 9.1** (the shape equation running until it fails). Take  $m = c = 1$ , so that  $C_n = 1 + n$  and the shape equation (5.1) reads  $D_n + 1 = (a_n - 1)(1 + n)$ , and take  $D_0 = 1$ . Each step is now determined: at  $n = 0$  the equation reads  $2 = (a_0 - 1) \cdot 1$ , so  $a_0 = 3$ , and  $D_1 = a_0 D_0 = 3$ ; at  $n = 1$  it reads  $4 = (a_1 - 1) \cdot 2$ , so  $a_1 = 3$ , and  $D_2 = a_1 D_1 = 9$ ; at  $n = 2$  it reads  $10 = (a_2 - 1) \cdot 3$ , which has no integer solution, and the orbit stops. Longer prefixes occur for other starting values: the finite search recorded in Appendix B found none longer than 17 among the initial states below 5000.

**Theorem 9.2** (no constant negative magnitude). *There is no pair of sequences  $a, D : \mathbb{N} \rightarrow \mathbb{N}$  with  $a_n \geq 2$  for all  $n$  satisfying  $D_{n+1} = a_n D_n$  and (5.1), for any  $m > 0$  and any  $c$ . The same holds if the shape only begins at some index.*

*Proof when  $c$  and  $m$  are coprime.* Assume first  $\gcd(c, m) = 1$ . Two facts about the multipliers do all the work, and the difficulty lies in the fact that they concern the same finite set of primes from opposite directions: the first makes every multiplier meet that set, the second lets each prime of the set meet at most one multiplier.

Every  $a_j$  shares a prime with  $m$ . Suppose  $\gcd(a_j, m) = 1$ . Then  $m$  is invertible modulo  $a_j$ , so some  $n$  has  $c + nm \equiv 0$ , that is  $a_j \mid C_n$ ; and  $a_j \mid D_n$  for every  $n > j$ , since  $D$  is

multiplicative with  $a_j$  among its factors. Choosing such an  $n$  beyond  $j$  and reading (5.1) modulo  $a_j$  gives  $a_j \mid m$ , contradicting  $\gcd(a_j, m) = 1$  and  $a_j \geq 2$ .

A prime divisor of  $m$  occurs in at most one  $a_j$ . Suppose  $p \mid m$  and  $p \mid a_j$ . For  $n > j$  we have  $p \mid D_n$ , so (5.1) gives  $(a_n - 1)C_n \equiv 0 \pmod{p}$ . Now  $C_n = c + nm \equiv c$ , and  $p \nmid c$  because  $p \mid m$  and  $\gcd(c, m) = 1$ ; hence  $p \mid a_n - 1$ , so  $p \nmid a_n$ . Thus  $p$  cannot divide any later multiplier.

The two facts are incompatible. Each of the infinitely many multipliers needs a prime divisor of  $m$ , and each prime divisor of  $m$  serves at most one multiplier, while  $m$  has only finitely many prime divisors.  $\square$

The case  $\gcd(c, m) = 1$  is the [scale-primitive exclusion](#). The special case  $m = c = 1$ , worked through in Example 9.1, where  $m$  has no prime divisors at all and the first fact is immediately contradictory, is recorded separately as the [normalised exclusion](#).

*Removing the scale.* For general  $c$ , put  $g = \gcd(c, m)$ . Equation (5.1) shows  $g \mid D_n$  for every  $n$ , so dividing  $D$ ,  $c$  and  $m$  by  $g$  leaves a system of the same shape with coprime data, which the previous case excludes.  $\square$

Formalised as the [exclusion at every scale](#), and the eventual form, obtained by shifting the orbit to the first constant index, as the [eventual exclusion](#).

Theorem 9.2 excludes a centred state that is eventually constant and negative, at every magnitude and every scale. It says nothing about a negative state whose magnitude changes, and the argument genuinely uses constancy: the finiteness of the set of prime divisors of  $m$  is what the pigeonhole uses, and a varying magnitude presents a fresh set at each index.

## 10 Excluding a periodic negative magnitude

The next case allows the magnitude to vary, provided it repeats. Write  $e_n = -E_n > 0$  for the magnitude, as in Section 4, so that  $C_{n+1} = C_n + e_n$ ; suppose  $e$  has period  $h$ , meaning  $e_{n+h} = e_n$  for every  $n$ , and that the state gains a fixed *drift*  $M > 0$  over one period, meaning  $C_{n+h} = C_n + M$  for every  $n$ .

At  $h = 1$  this says that the magnitude is constant and that  $M$  is its value, which is the situation of Section 9; Theorem 9.2 already excludes it, and does so without the extra hypothesis  $e_n < a_n$  imposed below. The new content is  $h \geq 2$ .

Section 9 could run its pigeonhole on the prime divisors of the single number  $m$ . Here the magnitude changes inside a period, and the number available to run it on is the drift  $M$ . Three lemmas replace the two facts of Section 9. The first is that multipliers persist: every  $a_j$  divides every strictly later denominator state, the [persistence of a multiplier](#). The second is a *lock*: a prime already dividing  $D$  and also dividing the current multiplier is forced into the next tail state, the [one-step lock](#), and once present in both  $D$  and  $C$  it stays in every later  $D$ ,  $C$  and magnitude, the [persistence of a lock](#). The third transports a lock backwards through the period: if a divisor of the drift  $M$  occurs in two multipliers, it is a common divisor of  $C_0$  and of every magnitude, the [repeated-divisor transport](#), using that both the period relation and the drift iterate over any number of periods, the [iterated period](#) and the [iterated drift](#).

**Theorem 10.1** (no periodic negative magnitude). *Let  $a, D, C, e : \mathbb{N} \rightarrow \mathbb{N}$  with  $a_n \geq 2$ ,  $e_n > 0$  and  $e_n < a_n$  for every  $n$ , satisfying*

$$D_{n+1} = a_n D_n, \quad C_{n+1} = C_n + e_n, \quad D_n + e_n = (a_n - 1)C_n,$$

*and suppose  $e_{n+h} = e_n$  and  $C_{n+h} = C_n + M$  for some  $h > 0$  and  $M > 0$ . This is impossible.*

*Proof.* Strong induction on the drift  $M$ . The difficulty lies in the fact that a repeated prime divisor of  $M$  need not contradict anything directly. It instead forces a common scale on the whole orbit, which can be divided out, and the induction runs on the drift that the division reduces.

Suppose first that no prime divisor of  $M$  is a common divisor of  $C_0$  and of every magnitude. Repeated-divisor transport then applies in the contrapositive: a prime divisor of  $M$  occurring in two of the multipliers would be exactly such a common divisor, so each prime divisor of  $M$  occurs in at most one multiplier. Against this, the argument of Section 9 run phase by phase (along a fixed residue class modulo  $h$  the magnitude is constant and  $C$  advances by  $M$  at each period, so the multipliers of that phase meet an arithmetic progression exactly as in Section 9) makes every multiplier share a prime with  $M$ . Since  $M$  has only finitely many prime divisors and there are infinitely many multipliers, pigeonhole gives a contradiction.

Otherwise some prime  $p$  divides  $M$ , divides  $C_0$ , and divides every magnitude. Then  $p \mid C_n$  for every  $n$ , since  $C_{n+1} = C_n + e_n$  and both summands on the right are divisible by  $p$ , and the shape equation  $D_n + e_n = (a_n - 1)C_n$  then gives  $p \mid D_n$  for every  $n$ . Divide  $D$ ,  $C$  and  $e$  by  $p$ . The multipliers are untouched, so  $a_n \geq 2$  and  $e_n < a_n$  persist and the magnitudes stay positive; the three recurrences are homogeneous in  $(D, C, e)$  and so survive; the period is still  $h$ ; and the drift becomes  $M/p < M$ . The inductive hypothesis applies.  $\square$

The first of the two cases above is the [phase-primitive exclusion](#), the induction is the [exclusion at every scale](#), and the eventual form is the [eventual exclusion](#).

The bound  $e_n < a_n$  is a genuine restriction and not a normalisation. It is the range in which a multiplier cannot divide its own phase magnitude, which is what the lock argument needs. It holds in the intended reading, where  $e_n$  is a centred representative and  $a_n$  grows quadratically, but it is assumed here and not derived. The drift hypothesis  $M > 0$  is also used: a periodic magnitude with zero drift would be identically zero, which is the case already settled by Section 8.

*Remark.* On the canonical orbit of [7], read through the dictionary of Section 3, the bound  $e_n < a_n$  holds eventually. Lemma 4(2) there gives the centring range  $-C_n/2 \leq E_n < C_n/2$ , so  $e_n = |E_n| \leq C_n/2$  at a negative index, and the proof of Lemma 4(3) gives  $C_{n+1} \leq \frac{3}{2}C_n$  [7, pp. 11–12], hence the uniform bound  $C_n \leq C_0(3/2)^n$ . Against this, summability of  $\sum 1/a_n$  forces  $a_n \rightarrow \infty$ , and the growth hypothesis  $a_n^2/a_{n+1} \rightarrow 1$  gives  $a_{n+1} \geq a_n^2/2$  for all large  $n$ , hence  $a_{n+1} \geq 2a_n$  once  $a_n \geq 4$ ; so the multipliers grow at least geometrically with ratio 2 and outgrow the bound on  $C_n$ , making  $e_n/a_n$  eventually small. Two caveats. Summability is used for this step, so the estimate is not available from the state recurrence alone; and the conclusion is eventual and not universal, whereas Theorem 10.1 imposes  $e_n < a_n$  at every index, so it is the eventual form of the exclusion

cited above that applies. Under the periodicity hypothesis of Theorem 10.1 the estimate is in any case unnecessary, since a periodic magnitude is bounded while the multipliers tend to infinity.

## 11 Bounded rise cannot remain coprime to fresh moduli

Periodicity is still a strong assumption. Removing it needs a different kind of obstruction, and the one used here is a counting argument about where an integer sequence with bounded upward steps must land.

**Lemma 11.1** (shifted blocks of consecutive multiples). *Let  $m_0, \dots, m_{B-1}$  be pairwise coprime and at least 2. For every bound there is a  $t$  beyond it with  $m_i \mid t + i$  for each  $i < B$ .*

*Proof.* Standard Chinese remaindering: solve  $t \equiv -i \pmod{m_i}$  simultaneously, then add multiples of  $\prod_i m_i$  to pass the bound.  $\square$

Formalised as the [shifted consecutive multiples](#).

**Example 11.2** (a block of three consecutive multiples). Take  $B = 3$  and  $(m_0, m_1, m_2) = (3, 4, 5)$ . The congruences  $t \equiv 0 \pmod{3}$ ,  $t \equiv 3 \pmod{4}$  and  $t \equiv 3 \pmod{5}$  hold exactly when  $t \equiv 3 \pmod{60}$ . At  $t = 3$  the block is 3, 4, 5 with  $3 \mid 3$ ,  $4 \mid 4$  and  $5 \mid 5$ ; at  $t = 63$  it is 63, 64, 65 with  $3 \mid 63$ ,  $4 \mid 64$  and  $5 \mid 65$ . A sequence of natural numbers that starts at 0, tends to infinity and rises by at most 3 at each step cannot step over the block  $\{3, 4, 5\}$ : it has a first value at least 3, that value is at most 5, and every member of  $\{3, 4, 5\}$  is divisible by one of the three moduli.

**Theorem 11.3** (bounded rise cannot remain coprime to fresh moduli). *Let  $u : \mathbb{N} \rightarrow \mathbb{N}$  tend to infinity with  $u_{n+1} \leq u_n + B$  for a fixed  $B > 0$ . Then  $u$  cannot remain coprime to infinitely many fresh pairwise coprime moduli: there is no family of pairwise coprime  $m_i \geq 2$ , one for each index, such that  $\gcd(m_i, u_i) = 1$  whenever  $i < t$ .*

*Proof.* Take  $B$  of the moduli, from indices beyond any chosen point, and use Lemma 11.1 to find  $t$  far out with the  $i$ th of them dividing  $t + i$ . The interval  $[t, t + B)$  has length  $B$ , and  $u$  climbs by at most  $B$  per step while tending to infinity, so some  $u_s$  lands in it. Then  $u_s = t + i$  for some  $i < B$ , and the  $i$ th modulus divides  $u_s$ . Since that modulus is at least 2, this contradicts  $\gcd(m_i, u_i) = 1$ .  $\square$

Formalised as the [bounded-rise barrier](#). The key point is that the two hypotheses pull against each other exactly: divergence forces  $u$  to travel arbitrarily far, and bounded rise forbids it from stepping over a window of width  $B$ . Uniformity of the bound is what the proof uses, and it is the hypothesis that cannot be relaxed by this argument: a rise bounded only along a subsequence leaves the intervening steps free to jump the window. We have not looked for a sequence of unbounded rise remaining coprime to every fresh modulus. The statement is elementary and we are not aware of a prior published form of it, but we would not be surprised if one exists.

The moduli are indexed by the same set as the sequence, and the  $i$ th of them is asked to divide no value  $u_t$  with  $t > i$ ; nothing at all is asked of it at or before its own index.

That asymmetry is what makes the hypothesis available in the application below, where  $m_i$  is the  $i$ th multiplier and only the later numerators are known to be coprime to it.

The barrier applies to the problem because a *reduced* exact tail already carries pairwise coprime moduli. Call  $(u, v)$  a reduced exact tail with multipliers  $a$  when  $\gcd(u_n, v_n) = 1$ ,

$$u_{n+1} + v_n = a_n u_n, \quad v_{n+1} = a_n v_n.$$

Here  $u$  is the numerator and  $v$  the denominator: the two recurrences are those of Section 4, with  $u$  in the role of the tail state  $C$  and  $v$  in that of the denominator state  $D$ , and with coprimality imposed at every index.

**Proposition 11.4** (reduced tails are pairwise coprime). *In a reduced exact tail,  $a_n$  is coprime to  $v_n$ ; the multipliers at distinct indices are pairwise coprime; and every earlier multiplier is coprime to every later numerator.*

These are the [step coprimality](#), the [pairwise coprimality](#), and the [whole-modulus avoidance](#). Feeding them to Theorem 11.3 gives the form used later: there is no reduced exact tail whose numerator tends to infinity with a uniformly bounded upward increment, the [reduced bounded-rise exclusion](#), together with its eventual form, the [eventual reduced exclusion](#).

**Example 11.5** (Sylvester's sequence as a reduced exact tail). Put  $u_n = 1$  and  $v_n = D_n = 1, 2, 6, 42, 1806, \dots$  as in Example 4.2, with multipliers  $a_n = v_n + 1 = 2, 3, 7, 43, 1807, \dots$ . Then  $\gcd(u_n, v_n) = 1$ ,  $u_{n+1} + v_n = 1 + v_n = a_n u_n$  and  $v_{n+1} = a_n v_n$ , so this is a reduced exact tail, and Proposition 11.4 returns the classical fact that Sylvester's numbers are pairwise coprime. Its numerator is constant, so it satisfies every hypothesis of the exclusion just stated except divergence; since the tail exists, that hypothesis cannot be dropped.

An arbitrary orbit of Section 4 need not be reduced, so one more step is needed before the barrier can be applied to it: the common factor must stop changing. We call a set of indices *cofinal* when it is infinite, and say that a property holds *cofinally* when the set of indices at which it holds is cofinal. The step below is the only one connecting the orbit of Section 4 to the reduced tails of Proposition 11.4, and it is where the cofinal boundedness in the proof of Theorem 12.1 is used.

**Proposition 11.6** (the tail gcd stabilises). *Let  $(a, D, C)$  be an exact natural orbit, that is, a triple of  $\mathbb{N}$ -valued sequences with  $C_{n+1} + D_n = a_n C_n$  and  $D_{n+1} = a_n D_n$ , whose centred state  $E_n = \text{ctr}(a_n, D_n, C_n)$  is negative cofinally, with magnitudes bounded along that cofinal set. Then  $\gcd(C_n, D_n)$  is eventually constant, and beyond that index the orbit divided by the stable gcd is a reduced exact tail.*

*Proof.* The tail gcd  $\gcd(C_n, D_n)$  divides its successor, the [gcd monotonicity](#), so the tail gcds form a positive divisibility chain. At an index where the centred state is negative the tail gcd is exactly  $\gcd(C_n, e_n)$ , the [gcd at a negative index](#), and is therefore at most the magnitude there; so the chain is bounded along the cofinal set of negative indices. A positive divisibility chain whose values are bounded along a cofinal set is eventually constant, the [chain stabilisation](#). Hence cofinally bounded negative magnitudes make

the tail gcd stabilise, the [gcd stabilisation](#), and beyond that point dividing by the stable gcd leaves the numerator and denominator coprime, that is, a reduced exact tail.  $\square$

The stabilisation of the tail gcd is the checked statement cited at the end of the proof; the last clause, that the divided orbit is a reduced exact tail, is exposition and carries no separate label. Boundedness is required only *along a cofinal set*. That is what the situation gives: the tail gcd is controlled only at the negative indices, and cofinally many of them is all the divisibility chain needs.

Normalised vanishing by itself gives a different, strictly weaker conclusion. For an exact natural orbit with  $C_n > 0$ , put

$$G_n = \gcd(C_n, D_n), \quad \Gamma(N) = \#\{0 \leq j < N : G_j < G_{j+1}\}.$$

**Proposition 11.7** (normalised vanishing makes strict gcd changes sparse). *Let  $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$  satisfy  $C_{n+1} + D_n = a_n C_n$  and  $D_{n+1} = a_n D_n$  with  $C_n > 0$ , and put  $E_n = D_n - (a_n - 1)C_n$ ,  $G_n = \gcd(C_n, D_n)$  and  $\Gamma(N) = \#\{0 \leq j < N : G_j < G_{j+1}\}$ . If  $|E_n|/C_n \rightarrow 0$ , then  $\Gamma(N) = o(N)$ . Moreover, for every starting bound  $B$  and block length  $L$ , some  $n \geq B$  satisfies*

$$G_n = G_{n+1} = \dots = G_{n+L}.$$

The first assertion is the [sublinear strict-growth theorem](#); the second is the [arbitrarily late constant-block theorem](#). The proof first converts normalised vanishing into subexponential growth of  $C_n$ , then charges every strict divisibility increase of  $G_n$  against a finite power-of-two budget. Sparse strict changes force long finite gaps between them.

The quantifiers matter. Proposition 11.6 uses cofinally bounded negative magnitudes and yields eventual constancy. Proposition 11.7 uses only normalised vanishing and yields arbitrarily late constant blocks of each prescribed finite length. It does not yield one infinite constant tail, and it does not exclude cofinally unbounded negative excursions.

## 12 Excluding a bounded negative part

Assembling the previous section gives Theorem 12.1, the endpoint of the barrier argument. Theorem 13.1 in the next section uses a different finiteness hypothesis. The two results are incomparable: Theorem 13.1 assumes only the state recurrence with  $C_n > 0$ , while hypothesis (5) below does not imply finite *normalised negative mass*, by which we mean convergence of  $\sum_n (-E_n)_+ / C_n$ , where  $x_+ = \max(x, 0)$ , since a constant negative part  $-b$  gives  $C_n \sim bn$  and hence a divergent sum. The hypotheses of Theorem 12.1 are listed in full, because two of them are analytic inputs that are assumed here rather than derived from the growth condition of Problem 1.1.

Informally, hypotheses (5) and (6) below pull against each other. By Proposition 4.1, hypothesis (5) caps the rise of the tail state at  $B$  per step, while hypothesis (6), once the error is nowhere zero, forces the tail state to diverge. Proposition 11.6 makes the orbit reduced, and Theorem 11.3 then converts the tension into a contradiction: an integer sequence tending to infinity with upward steps of size at most  $B$  cannot remain coprime to the pairwise coprime moduli a reduced tail carries. The complementary case, in which the error is eventually positive, is the descent of Theorem 8.1.

**Theorem 12.1** (bounded negative part). *Let  $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$  and  $E : \mathbb{N} \rightarrow \mathbb{Z}$  satisfy*

1.  $a_n > 1$  and  $C_n > 0$  for every  $n$ ;
2. the exact dynamics  $C_{n+1} + D_n = a_n C_n$  and  $D_{n+1} = a_n D_n$ ;
3.  $E_n = \text{ctr}(a_n, D_n, C_n)$  for every  $n$ ;
4. eventual strict centring:  $|E_n| < C_n$  for all large  $n$ ;
5. eventually bounded negative part:  $-B \leq E_n$  for all large  $n$ , for some  $B$ ;
6. normalised vanishing: for every  $K$  there is an  $N$  with  $K|E_n| < C_n$  for all  $n \geq N$ .

Then  $E_n = 0$  for all sufficiently large  $n$ .

*Proof.* Suppose not. We first remove every late zero: by Theorem 5.7 and its contrapositive form,  $E$  is nowhere zero beyond the centring threshold. Two cases remain. If  $E$  is negative cofinally, then (5) bounds those magnitudes, so Proposition 11.6 makes the tail gcd stabilise and the orbit may be taken reduced past that point. Hypothesis (5) with Proposition 4.1 gives  $C_{n+1} \leq C_n + B$ , a bounded rise; hypothesis (6) with  $E$  nowhere zero gives  $C_n \rightarrow \infty$ . A reduced exact tail with a divergent numerator and bounded rise is excluded by Theorem 11.3. It remains to treat the case that  $E$  is eventually positive, and there the descent of Theorem 8.1 applies and forces  $E$  to vanish, contradicting nowhere-vanishing.  $\square$

Formalised as the [bounded-negative-part rigidity](#), with the eventual form, in which the centring and the bound need only hold from some index, as the [eventual bounded-negative-part rigidity](#). The two auxiliary results it uses are the [divergence from normalised vanishing](#) and the [exclusion of cofinally bounded negative magnitudes](#), the latter over the [tail-divergence form](#).

**Corollary 12.2.** *Under the hypotheses of Theorem 12.1, together with  $C_{n+1} \neq 0$  for all large  $n$ , the multipliers satisfy  $a_{n+1} = a_n^2 - a_n + 1$  for all sufficiently large  $n$ .*

*Proof.* Theorem 12.1 then Theorem 5.6.  $\square$

Hypotheses (1)–(3) are the state system, and (6) is the division-free form of  $|E_n|/C_n \rightarrow 0$ . Hypothesis (4) is logically redundant: hypothesis (6) with  $K = 1$  gives  $|E_n| < C_n$  eventually. It is retained because the displayed statement tracks the linked Lean declaration; shifting to the resulting threshold gives the same conclusion without an independent centring input. The formal source does not derive normalised vanishing from  $a_{n+1}/a_n^2 \rightarrow 1$ , so Theorem 12.1 is, by itself, a conditional theorem about the state system. Combined with Koizumi’s bridge below, however, Corollary 12.2 settles the bounded-negative canonical case of Problem 1.1. What is unconditional in this note is the all-negative constant and periodic exclusions, Theorems 9.2 and 10.1, and the barrier itself, Theorem 11.3.

The conditionality can nevertheless be located precisely, because the missing inputs are supplied elsewhere. With the results of [7], Theorem 12.1 becomes a conditional theorem about Problem 1.1 itself rather than about the state system alone. Let  $(a_n)$  satisfy the hypotheses of Problem 1.1. After deleting a finite prefix, Corollary 3

of [7, p. 9] makes the sequence the pseudo-greedy expansion of its own reciprocal sum, and Lemma 4 there supplies the integers  $c_n, d_n, e_n$  of Section 3. Hypothesis (1) then holds:  $c_n$  is a positive integer by Lemma 4(1), and  $a_n > 1$  after a further finite shift, because summability of  $\sum 1/a_n$  forces  $a_n \rightarrow \infty$ . Hypothesis (2) is the pair of recurrences  $c_{n+1} = a_n c_n - d_n$  and  $d_{n+1} = a_n d_n$  of Lemma 4(2), and hypothesis (3) is that lemma's  $a_n = (d_n - e_n)/c_n + 1$  read as  $e_n = d_n - (a_n - 1)c_n$ , which is the map ctr. The centring range  $-c_n/2 \leq e_n < c_n/2$  in the same lemma gives hypothesis (4) at every index, which is stronger than the eventual form used here. Hypothesis (6) is the vanishing of the gap sequence in Corollary 3, since  $\varepsilon_n = e_n/c_n$ . The sole hypothesis not supplied is (5), the eventual bound on the negative part; nothing in [7] yields it and nothing here does either, so Problem #243 stays open.

### 13 Excluding finite normalised negative mass

Theorem 12.1 uses a uniform bound on the negative part. A different finiteness hypothesis, on the total normalised negative mass, removes a further family of orbits, and the argument is multiplicative rather than arithmetic: it compares  $C$  with a convergent product instead of with a system of congruences.

**Theorem 13.1** (finite normalised negative mass). *Let  $C_n \in \mathbb{N}_{>0}$  and  $E_n \in \mathbb{Z}$  satisfy  $C_{n+1} = C_n - E_n$ . If*

$$\sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} < \infty,$$

*then  $E_n = 0$  eventually. For an exact reciprocal-tail orbit this implies  $a_{n+1} = a_n^2 - a_n + 1$  eventually. No normalised-vanishing hypothesis is required.*

*Proof.* Put  $\delta_n = (-E_n)_+/C_n$ . The update gives  $C_{n+1} \leq C_n(1 + \delta_n)$ , so

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n) \leq C_0 \exp\left(\sum_n \delta_n\right).$$

Choose an integer upper bound  $K$  for  $C_n$ . Each strict rise of the integer sequence contributes at least  $1/K$  to  $\sum_n \delta_n$ , so there are only finitely many rises. The remaining positive integer sequence is nonincreasing and therefore stabilises, forcing  $E_n = 0$ . The recurrence follows from Theorem 5.6, since  $C_n > 0$ .  $\square$

The scalar conclusion is the [eventual-vanishing theorem](#); its [exact-orbit consequence](#) gives the Sylvester recurrence. For positive integer denominators with a rational reciprocal sum, the [canonical finite-mass criterion](#) supplies the exact state and its positivity. Summability of the negative mass remains an assumption; no growth or normalised-vanishing hypothesis is needed for this implication.

The transfer of the previous section applies here as well. On the canonical orbit of Corollary 3 of [7, p. 9] the recurrence  $C_{n+1} = C_n - E_n$  holds with  $C_n > 0$  by Lemma 4 [7, pp. 11–12]. Thus Theorem 13.1 is a conditional theorem about Problem 1.1 whose sole remaining hypothesis is summability of  $\sum_n (-E_n)_+/C_n$ . That hypothesis is not supplied there, and is the content of Problem 14.5 below.

**Example 13.2** (the product bound at a geometric rate). Suppose  $C_0 = 100$  and  $(-E_n)_+/C_n \leq 2^{-n-1}$  for every  $n$ , so that the normalised negative mass is at most 1; the constraint at  $n = 0$  still permits  $E_0$  as negative as  $-50$ . Since  $1 + x \leq e^x$ ,

$$C_N \leq 100 \prod_{n < N} (1 + 2^{-n-1}) \leq 100e < 272$$

for every  $N$ . For all sufficiently large  $n$ , we have  $2^{-n-1} < 1/272$ . A strict rise would then force  $(-E_n)_+/C_n \geq 1/C_n > 1/272$ , a contradiction. Thus  $C_n$  is eventually nonincreasing; as a positive integer sequence it stabilises, and  $E_n = 0$  thereafter. The constant 272 comes from the total mass and not from any single magnitude, which is why the hypothesis of Theorem 13.1 is summability of  $(-E_n)_+/C_n$  and not a bound on it.

## 14 Complements and further questions

Problem #243 is open. The negative case has narrowed, and it is worth being precise about what is left.

Each exclusion holds only in its stated regime. Theorems 9.2 and 10.1 exclude the constant and the stated periodic magnitudes on an all-negative tail, and do not exclude those patterns merely along the negative indices of a mixed-sign tail; Theorems 12.1 and 13.1 exclude a bounded negative part and finite normalised negative mass under normalised vanishing. Together with absorption and descent they fix the profile of any counterexample.

**Proposition 14.1** (frontier profile for a counterexample). *For the canonical integer state attached to any counterexample to Problem 1.1,*

$$E_n \neq 0 \text{ eventually,} \quad \frac{|E_n|}{C_n} \rightarrow 0,$$

and

$$\limsup_{\substack{n \rightarrow \infty \\ E_n < 0}} (-E_n) = \infty, \quad \sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} = \infty.$$

*In particular, negative indices occur infinitely often. Thus the remaining regime consists of globally coherent, unbounded negative excursions with divergent normalised negative mass.*

*Proof.* Koizumi's canonical-tail transfer gives strict centring and normalised vanishing after a finite shift; the strict-centring hypothesis of Theorem 12.1 also follows from normalised vanishing with  $K = 1$ . Absorption then makes  $E_n \neq 0$  eventually, since eventual zero would give the Sylvester recurrence. Descent excludes an eventually nonnegative state. Theorem 12.1 excludes a bounded negative part, and Theorem 13.1 excludes finite normalised negative mass. These statements are unchanged by deleting a finite prefix.  $\square$

Put

$$\delta_n := \frac{(-E_n)_+}{C_n}.$$

Every argument in Sections 9–12 uses a finiteness hypothesis: a fixed set of prime divisors, a fixed period, or a fixed bound on the negative part. Theorem 13.1 uses  $\sum_n \delta_n < \infty$ . Proposition 14.1 records that a counterexample has none of them; it is a frontier statement, not an additional problem equivalent to the original one. Stated at the level of state orbits, the surviving obstruction is the following.

**Problem 14.2** (unbounded divergent-mass negative excursions). Exclude, or construct, an exact natural orbit  $(a, D, C)$  with  $a_n > 1$  and  $C_n > 0$ , whose multipliers satisfy  $\lim_n a_{n+1}/a_n^2 = 1$  and whose centred state satisfies normalised vanishing, and which is negative infinitely often with magnitudes unbounded along that cofinal set and

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

An excursion of this kind has none of the finiteness hypotheses just listed. By Proposition 14.1 the canonical state of a counterexample is such an orbit after deletion of a finite prefix, so an exclusion would settle Problem #243. A proposed construction must be global, and it must supply the growth condition of Problem 1.1 together with a vanishing tail ratio, under which the [reciprocal-series realisation](#) makes the orbit the canonical orbit of the rational number  $C_0/D_0$ . Finite admissible prefixes alone do not provide such a construction.

*Remark.* The hypotheses in Problem 14.2 beyond the state recurrence are not decorative: the recurrence alone admits unbounded divergent-mass excursions. Put

$$C_n = 2^n, \quad b_0 = 2, \quad b_{n+1} = \frac{1}{2}b_n(b_n + 2), \quad a_n = b_n + 2, \quad D_n = b_n C_n,$$

so that  $(b_n) = 2, 4, 12, 84, 3612, \dots$  and  $(a_n) = 4, 6, 14, 86, 3614, \dots$ . Every  $b_n$  is a positive even integer, since  $b_n = 2k$  gives  $b_{n+1} = 2k(k+1)$ , so all four sequences are  $\mathbb{N}$ -valued. The orbit is exact:  $C_{n+1} + D_n = 2^n(2 + b_n) = a_n C_n$  and  $D_{n+1} = b_{n+1} 2^{n+1} = b_n(b_n + 2) 2^n = a_n D_n$ . Its centred state is

$$E_n = D_n - (a_n - 1)C_n = b_n 2^n - (b_n + 1)2^n = -C_n,$$

so the negative magnitudes  $2^n$  are unbounded and  $\sum_n (-E_n)_+/C_n$  diverges. What fails is the centring:  $|E_n| = C_n$  at every index, so strict centring fails at the boundary and normalised vanishing fails outright. The multipliers satisfy  $a_{n+1} = \frac{1}{2}(a_n^2 - 2a_n + 4)$ , so  $a_{n+1}/a_n^2 \rightarrow \frac{1}{2}$  and the growth hypothesis fails as well. The example therefore says nothing about Problem 1.1, and it does not isolate the centring hypotheses:  $E_n = -C_n$  at every index forces that multiplier recurrence, so growth fails with them. It is recorded only to show that the state recurrence alone is not enough.

## 14.1 Global height and old-factor overlap

The main arithmetic gap is global. Define the cumulative digit LCM and its overlap quotient by

$$L_0 = D_0, \quad L_{n+1} = \text{lcm}(L_n, a_n), \quad M_n = \frac{D_n}{L_n}.$$

Since  $D_n = D_0 \prod_{j < n} a_j$ , the quotient is integral and  $M_n L_n = D_n$ . It records the multiplicity lost when the full product scale is compressed to an LCM.

## 14.2 Repair entropy across a recovery

The cumulative quotient does not by itself distinguish one large deletion from many small ones. The reduced recurrence does. Let  $u, h, c : \mathbb{N} \rightarrow \mathbb{N}$  and  $e : \mathbb{N} \rightarrow \mathbb{Z}$  satisfy

$$h_n u_{n+1} = u_n - e_n, \quad u_n > 0, \quad h_n > 0.$$

Fix  $K, r, L \in \mathbb{N}$  with  $K, L > 0$ , assume

$$K|e_{r+i}| < u_{r+i} \quad (0 \leq i < L), \quad u_r \leq u_{r+L},$$

and let  $R$  be a finite family of moduli. Suppose every  $m_q, q \in R$ , divides the deletion product  $\prod_{r \leq n < r+L} c_n$ , and that  $c_n^2 \mid h_n$  throughout the interval. Then

$$K^L \text{lcm}(m_q : q \in R)^2 < (K+1)^L. \quad (22)$$

This is the [repair-entropy conservation law](#).

The square in (22) is the key point. Exact denominator reduction charges  $c_n^2$  to the scale growth, so a repaired family whose LCM is at least  $2^{|R|}$  satisfies

$$K^L 4^{|R|} < (K+1)^L.$$

The formal consequence is [the independent-repair cardinality bound](#). It rules out positive-density deletion of independent old moduli inside recoveries with a fixed relative-error budget.

There is also an endpoint that does not mention a chosen modulus family. If  $K|e_n| < u_n$  eventually for every  $K$ , then for each fixed  $L > 0$  there is an  $N$  such that every recovery  $u_r \leq u_{r+L}$  with  $r \geq N$  satisfies

$$\prod_{0 \leq i < L} h_{r+i} = 1.$$

Thus a surviving nontrivial reset must have recovery length tending to infinity. Lean checks this as [eventual disappearance of fixed-length reset payments](#). The theorem does not bound a recovery length that varies with  $r$ , nor does it supply a lower bound for the LCM of the moduli repaired in such an interval. Those are the missing inputs needed to turn (22) into a global contradiction.

**Problem 14.3** (global overlap-height growth). For every nonterminal canonical orbit satisfying Problem 1.1, must

$$\limsup_{n \rightarrow \infty} \frac{\log M_n}{n} > 0?$$

Equivalently, must there be a  $K \geq 1$  for which

$$2^n \leq M_n^K$$

at infinitely many indices?

The two displayed formulations are equivalent up to changing the positive constant; the second is not a weaker target. On the canonical orbit, Section 6 supplies  $M_n \mid C_n$  and  $M_n \leq C_n$ . Therefore a positive answer would contradict the subexponential tail-height budget and prove Problem #243. The missing input is the proposed lower bound on overlap growth.

There is also a more local-looking question whose content is nevertheless the entire prefix. Put  $A_n = \prod_{j < n} a_j$ , so  $D_n = D_0 A_n$ . From

$$D_0 A_n = (a_n - 1)C_n + E_n$$

one immediately obtains

$$\gcd(A_n, a_n - 1) \mid E_n.$$

The checked [tail-height estimate](#) and normalised vanishing make  $|E_n|$  subexponential in  $n$ .

**Problem 14.4** (old-factor overlap). In every nonterminal canonical orbit satisfying Problem 1.1, is

$$\limsup_{n \rightarrow \infty} \frac{\log \gcd(A_n, a_n - 1)}{n} > 0?$$

Equivalently, do there exist  $\eta > 0$  and infinitely many  $n$  such that  $\gcd(A_n, a_n - 1) \geq e^{\eta n}$ ?

A positive answer contradicts the displayed divisibility and the subexponential error budget. Arbitrarily long locally admissible blocks do not answer this question: the gcd uses the complete prefix.

### 14.3 The direct analytic question

**Problem 14.5** (summability from rationality). Let  $(a_n)$  satisfy Problem 1.1, and let  $(D_n, C_n, E_n)$  be its canonical integer state. Must

$$\sum_{n=0}^{\infty} \delta_n = \sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} < \infty?$$

An affirmative answer closes Problem #243 by Theorem 13.1. Since  $\delta_n \rightarrow 0$ , convergence is equivalently expressible as boundedness of the partial products  $\prod_{n < N} (1 + \delta_n)$ ; this is the same criterion in multiplicative form. A negative answer requires a sequence satisfying the full growth and rationality hypotheses. A locally admissible state orbit is insufficient. By Theorem 6.2, it is enough to establish the finiteness of (20) for one decreasing weight with divergent integral and one fixed baseline. This replaces the all-index mass by a discounted series supported only at LCM records.

*Remark.* One further question is not left open here, since it is answered in [7]: whether normalised vanishing follows from  $a_{n+1}/a_n^2 \rightarrow 1$  and rationality. It does. After deleting a finite prefix, Koizumi's Corollary 3 [7, p. 9] makes the sequence the pseudo-greedy expansion of its own reciprocal sum and gives  $\varepsilon_n \rightarrow 0$  under exactly the hypotheses of Problem 1.1; rationality is not needed for that step, only summability of  $\sum 1/a_n$ . Since  $\varepsilon_n = E_n/C_n$  under the intended reading of Section 4, this is normalised vanishing. Strict

centring comes from the same source and is stronger than what Theorem 12.1 assumes: for a rational reciprocal sum, the range  $-c_n/2 \leq e_n < c_n/2$  of his Lemma 4(2) [7, pp. 11–12] gives  $|E_n| < C_n$  at every index, which is stronger than eventual strict centring. The eventual form also follows directly from normalised vanishing with  $K = 1$ . What remains unsupplied is hypothesis (5), the bound on the negative part, which is why the theorem is still conditional.

#### 14.4 A barrier extension

**Problem 14.6** (variable-rise CRT barrier). Determine the weakest growth condition on the positive increments of  $u_n$  under which an unbounded integer sequence cannot remain coprime to infinitely many pairwise-coprime old moduli as in Theorem 11.3. In particular, does the conclusion remain valid under the candidate condition

$$(u_{n+1} - u_n)_+ = o(\log \log(u_n + 3))?$$

The displayed rate is a proposed threshold, not a proved sharp boundary. Either an extension of the CRT barrier or a counterexample with this rise rate would be informative independently of Problem #243. Proposition 7.12 bounds the search from above: under whole-modulus avoidance alone the conclusion already fails at the rate  $(1 + \epsilon)\ell(u_n)$ , so the answer lies in the band between  $o(\ell(u_n))$  and  $(1 + \epsilon)\ell(u_n)$ , and an affirmative answer must use coprimality itself rather than avoidance of multiples.

**Problem 14.7** (the maximal-gap coefficient). Let  $m_0 < m_1 < \dots$  be pairwise coprime with  $\ell(m_j) = j + O(1)$  and  $\sigma = \prod_j (1 - 1/m_j) > 0$ , and let  $u_n$  enumerate the integers divisible by no  $m_j$ . Is  $\limsup_n (u_{n+1} - u_n)/\ell(u_n) = \sigma^{-1}$ ? For the Fermat moduli  $m_j = 2^{2^j} + 1$  the product telescopes to  $\sigma = 1/2$  exactly, and the observed ratio is 1.8116 for  $u \geq 1000$  up to  $3 \cdot 10^6$  against the proposed value 2.

The upper bound implicit in Proposition 7.12 is  $(1 - \theta)^{-1}$  with  $\theta = \sum_j 1/m_j$ , which for the Fermat family is 2.475 ...; the proposed constant  $\sigma^{-1}$  is smaller, and no proof of it is offered here.

#### Formalisation targets

**Erdős–Straus.** Formalise Theorem 3 of [1], with its prefix-LCM quotient and correctly indexed growth factor, under the published analytic hypotheses.

**Duverney.** Formalise Corollary 3.2 of [2], including its signed numerators and its one-sided summability hypothesis, and then recover the all-positive specialisation used here. These are useful verification projects, but neither is an additional open mathematical problem.

#### Statements and declarations

**Artefact and data availability.** The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. Proof authority remains with those sources; this manuscript provides navigation and exposition.

Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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## A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 92b88dc1bbe0. The state system, the exclusions, the barrier, and the bounded-negative-part theorem are in `ReciprocalTailRigidity.lean`; the finite-negative-mass theorem is in `SparseResetRecovery.lean`; and the repair-entropy inequalities are in `RepairEntropy.lean`. The residue reduction of Appendix B is in `FiniteHorizon-Residue.lean`. Three distinctions are worth carrying into the source. The identification of any of these modules with reciprocal tails is the exposition of Section 4 and is not a checked statement. The periodic exclusion assumes the regime  $e_n < a_n$ . And the final Lean declaration lists strict centring and normalised vanishing as hypotheses; the former follows eventually from the latter with  $K = 1$ , while the formal source derives neither from the original analytic problem. For the external bridge supplying them, see Sections 3 and 12.

## B A factorial residue reduction for forced orbits

In the case  $m = c = 1$  of Section 9, where the shape equation (5.1) reads  $D_n + 1 = (a_n - 1)(n + 1)$ , each multiplier is determined by its predecessor; we call such an orbit *forced*. At index  $n$  the numerator of the next multiplier is

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the [forced numerator](#), and the divisor is  $n + 2$ . The orbit survives a step when that division is exact, giving a survival predicate, the [survival predicate](#). Example 9.1 is the forced orbit from  $a = 3$  read this way:  $\text{num}(0, 3) = 6$  is divisible by 2 and gives  $a_1 = 3$ , while  $\text{num}(1, 3) = 13$  is not divisible by 3, so the orbit stops there. Deciding survival by iteration is expensive because the orbit grows doubly exponentially, and it is unnecessary: survival over a finite horizon depends on the initial value only through a factorial residue.

**Theorem B.1** (factorial residue reduction). *For all  $h$  and all integers  $a \equiv b \pmod{(h + 1)!}$ , the orbit from  $a$  survives  $h$  forced updates exactly when the orbit from  $b$  does.*

*Proof.* Let  $M(0, i) = 1$  and  $M(h + 1, i) = (i + 2)M(h, i + 1)$ , an ascending factorial with  $M(h, 0) = (h + 1)!$ . The numerator is a polynomial with integer coefficients, so congruences transfer; reducing the modulus gives divisibility by  $i + 2$  for one exactly when for

the other, and cancelling that common factor from both values and modulus leaves the inductive hypothesis at  $M(h, i + 1)$ .  $\square$

Formalised as the [factorial residue reduction](#), over the [shrinking-modulus transport](#), the [polynomial congruence](#), and the [exact-division cancellation](#); the modulus identifications are the [ascending-factorial form](#) and the [factorial value at the initial index](#).

At  $h = 1$  the modulus is  $2! = 2$ , and surviving one update means  $2 \mid a^2 - 2a + 3$ , which holds exactly for odd  $a$ : for instance  $\text{num}(0, 3) = 6$  but  $\text{num}(0, 4) = 11$ . So one step of survival is decided by the parity of  $a$  alone, which is Theorem [B.1](#) at its smallest nontrivial horizon.

The search this supported is superseded. Running it over initial states below 5000 produced forced prefixes of length 17 and no longer; Theorem [9.2](#) now excludes the constant-negative case outright, for every seed and at every scale. The reduction is retained because it is exact, and because the shrinking-modulus technique transfers to any forced orbit whose step is a polynomial division.

## C Complete result-family map

This section places every registered family for this problem in the shared 70-family reader order. The five display bands control exposition only; the separate promotion state currently covers 6 families and is reported but does not hide or strengthen any family. Mathematical statements and evidence modes come from the public claim registry. Across all eight problems the public result-atom catalog contains 681 exact packet coordinates; this problem contributes 83. The recovered source catalog contains 682 rows; 1 row(s) belong to families absent from the current claim registry and remain disclosed as detached source rows. Catalog rows expose bounded statement excerpts plus full-source digests, not a claim that every complete packet statement is reproduced here.

### C.1 Centered state dynamics

**Reader position.** 2 of 70; display band: front door. Formal editorial disposition: split. These are separate classifications.

**Reader entry.** Under exact natural C/D dynamics with  $a > 1$ ,  $C > 0$ ,  $E = D - (a - 1)C$ , strict  $|E| < C$ , a uniform lower bound on  $E$ , and normalized vanishing, `boundedNegativePart_eventually_zero` forces  $E = 0$  eventually. `centered-State_eventually_zero` supplies the nonnegative stabilization mechanism, and `syvesterNext_eventually_of_centered_zero` recovers eventual Sylvester recurrence when the next tail is nonzero.

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**Authority and reach.** Lean kernel; Comparator-selected; locally proved signed recovery/rigidity theorem; novelty unassessed.

**Exact boundary.** The signed theorem has a  $C D : \mathbb{N} \rightarrow \mathbb{N}$ ,  $E : \mathbb{N} \rightarrow \mathbb{Z}$ , and  $B : \mathbb{N}$  with all exact hypotheses explicit:  $a > 1$ ,  $C > 0$ ,  $C(n + 1) + D(n) = a(n) \cdot C(n)$ ,  $D(n + 1) = a(n) \cdot D(n)$ ,  $E$  is the centered state  $D - (a - 1)C$ ,  $|E| < C$ , a uniform lower bound on  $E$ , and division-free normalized vanishing. It excludes persistent bounded negative behaviour and forces eventual centered defect zero; centered-zero plus an eventually nonzero next tail recovers Sylvester, but no reciprocal-tail irrationality follows. The unbounded negative mixed-sign branch and the prime-specific producer remain open. This is distinct from the periodic negative-orbit and cofinally bounded-negative no-go families; no novelty, priority, significance, or external-review claim is made, and no unrestricted #243 solution follows.

**Result-atom population.** 32 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

- `ErdosProblems.Erdos243.boundedNegativePart_eventually_zero`
- `ErdosProblems.Erdos243.centeredState_eventually_zero`
- `ErdosProblems.Erdos243.sylvesterNext_eventually_of_centered_zero`

## C.2 Negative mass recovery

**Reader position.** 10 of 70; display band: major result. Formal editorial disposition: promote. These are separate classifications.

**Reader entry.** Under exact reciprocal-tail C/D dynamics, positive tail, the strict centered-step equation, and division-free normalized vanishing, summability of the normalized negative centered mass forces `centeredState` to vanish eventually and recovers the Sylvester recurrence. The load-bearing theorem is supported by the exact tail-growth bound and the finite-negative-mass stabilization theorem.

Under exact reciprocal-tail C/D dynamics, positive tail, the strict centered-step equation, and division-free normalized vanishing, summability of the normalized negative centered mass forces `centeredState` to vanish eventually and recovers the Sylvester recurrence. The load-bearing theorem is supported by the exact tail-growth bound and the finite-negative-mass stabilization theorem.

**Authority and reach.** Lean kernel; conditional analytic recovery criterion; novelty unassessed.

**Exact boundary.** The exact criterion assumes  $a, D : \mathbb{N} \rightarrow \mathbb{Z}$  and  $C : \mathbb{N} \rightarrow \mathbb{N}$  with the declared next-denominator and next-tail dynamics,  $C > 0$ , the strict centered step equation, division-free normalized vanishing, and `Summable (negativeRelativeMass C (fun n ↦ centeredState (a n) (D n) (C n)))`. It does not prove these hypotheses for the original Erdős #243 orbit, supply a prime-specific producer, or establish an unconditional #243 endpoint or reciprocal-tail irrationality. Any surviving canonical orbit must have divergent normalized negative mass. This is distinct from `centered_state_dynamics`, whose recovery criterion uses a uniform lower-bound hypothesis; no novelty, priority, significance, or external-review claim is made.

**Result-atom population.** 36 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

- `ErdosProblems.Erdos243.tail_growth_le_one_add_negativeRelativeMass`
- `ErdosProblems.Erdos243.eventually_zero_of_summable_negativeRelativeMass`
- `ErdosProblems.Erdos243.sylvesterNext_eventually_of_summable_negativeRelativeMass`

## C.3 Bounded negative exclusion

**Reader position.** 15 of 70; display band: major result. Formal editorial disposition: promote. These are separate classifications.

**Reader entry.** Normalised vanishing excludes a cofinally bounded negative part and forces eventual Sylvester behaviour.

Normalised vanishing excludes a cofinally bounded negative part and forces eventual Sylvester behaviour.

**Authority and reach.** Lean kernel; Comparator-selected; conditional reduction and no-go result.

**Exact boundary.** Every positivity, dynamics, bounded-rise, and vanishing hypothesis remains explicit.

**Result-atom population.** 2 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

#### C.4 Bounded rise coprimality

**Reader position.** 26 of 70; display band: mechanism. Formal editorial disposition: merge. These are separate classifications.

**Reader entry.** A bounded-rise sequence cannot remain coprime to fresh pairwise-coprime moduli; reduced tails inherit this obstruction.

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**Authority and reach.** Lean kernel; locally proved result; novelty unassessed.

**Exact boundary.** The required bounded-rise input is not automatic for the original sequence.

**Result-atom population.** 11 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** represented by selected interface; 0 executable interface(s), 0 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

- ErdosProblems.Erdos243.no\_eventuallyBoundedRise\_reducedTail
- ErdosProblems.Erdos243.reducedTail\_pairwiseCoprime

#### C.5 Negative orbit no go

**Reader position.** 49 of 70; display band: frontier. Formal editorial disposition: promote. These are separate classifications.

**Reader entry.** Constant, eventually constant, periodic, and eventually periodic negative-error orbits are excluded in their stated regimes.

Constant, eventually constant, periodic, and eventually periodic negative-error orbits are excluded in their stated regimes.

**Authority and reach.** Lean kernel; Comparator-selected; no-go result.

**Exact boundary.** The exclusions do not cover arbitrary unbounded negative behaviour.

**Result-atom population.** 2 of 681 public coordinates. The public atom catalog groups them under this family.

**Comparator assurance.** exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

**Publication placement.** This family is also admitted to the short note.

## References

- [1] P. Erdős and E. G. Straus, *On the irrationality of certain Ahmes series*, J. Indian Math. Soc. (N.S.) **27** (1964), 129–133. MR 175848.
- [2] D. Duverney, *Irrationality of fast converging series of rational numbers*, J. Math. Sci. Univ. Tokyo **8** (2001), 275–316. MR 1837165.
- [3] C. Badea, *A theorem on irrationality of infinite series and applications*, Acta Arith. **63** (1993), no. 4, 313–323.
- [4] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. **28**, Geneva, 1980, p. 64.
- [5] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.
- [6] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. **175** (2025), 572–608; preprint arXiv:2406.17593v4.
- [7] J. Koizumi, *Irrationality of the reciprocal sum of doubly exponential sequences*, Integers **26** (2026), Paper No. A28, 17 pp., doi:10.5281/zenodo.18714404; preprint arXiv:2504.05933v1. Locators follow the published article; the arXiv v1 crosswalk is recorded in the source-closure note.
- [8] T. F. Bloom, *Erdős Problem #243*, erdosproblems.com/243, accessed 28 July 2026 (page displays “last edited 21 January 2026”). The current record labels the problem open, cites [ErGr80, p. 64] and [Er88c, p. 105], and explicitly describes its status as the website owner’s present assessment rather than a literature-completeness guarantee.
- [9] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.243*, Lean source at commit f776d2f, 2025, accessed 28 July 2026. Its  $\mathbb{Q}$ -valued Summable hypothesis encodes rationality, its indexing is zero-based, and its proof ends in sorry.